

Informative title: Depression and neuroticism decrease among women but not among men between 1976-2016 in Swedish septuagenarians

Running title: Depression and neuroticism decrease among women

Post-print submission 2019-01-17

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Acknowledgement

We would like to thank all study participants and research staff in the Gothenburg H70 Birth Cohort Studies to have made this work possible. Support for this work was provided by grants from The Swedish Research Council (11267, 825-2007-7462, 825-2012-5041, 521-2013-2699, RAM 2013-8717, 2015-02830, 2016-01590) , Swedish Research Council for Health, Working Life and Welfare (no 2001-2835, 2004-0145, 2006-0596, 2008-1111, 2008-1229, 2010-0870, 2012-1138, 2013-1202, 2016-2016-07097, AGECAP 2013-2300, 2013-2496, 2013-0475, Epilife 2006-1506), Sahlgrenska University Hospital (ALF), Gun & Bertil Stohnes forskningsstipendium, Fredrik & Ingrid Thuring's stiftelse, Hjalmar Svensson

Foundation, Konung Gustav och Drottning Victorias Frimurarstiftelsen, Söderström-König
Foundation and Stiftelsen Ragnhild & Einar Lundströms minne. This study was accomplished
while first author Therese Rydberg Sterner was affiliated with the Swedish National Graduate
School for Competitive Science on Ageing and Health (SWEAH), which is funded by the
Swedish Research Council.

Conflict of interest

None.

Description of authors' contributions

Resources and supervision: IS (senior author), MW. Formal analyses: TRS, NS.
Writing – original draft preparation: TRS, IS. Writing – review and editing: TRS, PG, RS,
FA, NS, AZ, SK, SÖ, MW, IS.

Data Accessibility Statement

Any data not published and supporting the results within the article are available from the
corresponding author upon reasonable request.

ABSTRACT

Objectives: We evaluated birth cohort differences in depressive symptom burden, prevalence of depression diagnoses, and neuroticism, among Swedish 70-year-olds examined between 1976 and 2016.

Methods: We used a repeated cross-sectional design examining four representative population samples of Swedish 70-year-olds (total n=2279) with identical methods in 1976-77 (n=392), 1992-93 (n=226), 2000-02 (n=487), and 2014-16 (n=1166). Depressive symptom burden was rated with the Montgomery Åsberg Depression Rating Scale. Major depression was diagnosed according to DSM-5, and minor depression according to DSM-IV-TR research criteria. Neuroticism was rated with the Eysenck Personality Inventory.

Results: For women in 2014-16, MADRS score (4.4 vs. 6.1 vs. 5.8; $p<0.05$) and neuroticism (6.6 vs. 7.7 vs. 9.2; $p<0.05$) were lower compared to 1992-93 and 1976-77, and the prevalence of any depression was lower compared to 2000-02 and 1992-93 (10.9% vs. 16.9% vs. 18.1%; $p<0.05$). For men, we observed no birth cohort differences in depression, while neuroticism was found to be lower in 2014-16 compared to 1976-77 among men without depression (5.1 vs. 5.9; $p<0.01$). The sex difference for MADRS and neuroticism declined between 1976-77 and 2014-16 (cohort*sex $p<0.05$).

Conclusions: Depressive burden and neuroticism decreased in 70-year-old women between 1976 and 2016.

Keywords: Depression; epidemiology; gender; old age; public mental health

Significant Outcomes:

- Projections for depression rates in the population are uncertain as the prevalence may change over time and may differ between birth cohorts. Our results show fluctuating prevalence between 1976 and 2016, however a decrease only statistically significant for women.
- Our findings show a decreased sex ratio in depression and neuroticism between 1976 and 2016, suggesting that environmental factors may play a role when contradicting time trend results for men and women are reported among studies.

Limitations:

- The 1922 birth cohort only includes women, which limited our ability to investigate time trends in men to the same extent.
- Some sub-groups in our analyses (e.g. men and women with major depression) were small, which limited the statistical power and may have generated false negative results.

INTRODUCTION

Depression is one of the most common mental disorders in old age ¹, and the leading cause of global burden of disease in both men and women ². Although prevalence estimates vary among studies ³, a consistent finding is that the prevalence of depression is about twice as high among women compared to men ⁴. Among older populations (> 65 years), the prevalence of depression is approximately 10% ⁵, including 4-5% with major depression ¹. According to the population-based Swedish National Study on Aging and Care in Kungsholmen (SNAC-K), the prevalence of depression is 5.9% among persons above age 60 (major depression: 0.8%, minor depression: 5.1%) ⁶, while the Survey of Health, Ageing and Retirement in Europe (SHARE) report a prevalence of 19.9 % among persons (>50 years) living in Sweden ⁷. Compared to younger age groups, depression seems to have stronger association with suicide in older persons ⁸.

As populations are ageing world-wide ⁹, most of the projected gains in life expectancy will occur among those above age 65 years ¹⁰, making late life depression an escalating public health problem. The global prevalence of major depression for all ages and both sexes combined has been shown to increase ¹¹. However, age- and sex-specific projections are uncertain as the prevalence of depression may change over time ¹². An increase in depression is suggested for younger ¹³, and middle-aged ¹⁴ populations. Among older adults the prevalence of major depression has been found to be stable ¹⁵, while milder forms of depression have been reported to increase ¹⁶, decrease ¹² or fluctuate ¹⁷ over time. Studies also show inconclusive time trend results ¹⁸. Few studies have examined time trends beyond 2010.

Neuroticism is a personality factor that is strongly related to depression, especially in late life ¹⁹. It is also more common in women compared to men ²⁰. Potential time trends in

neuroticism may thus influence the prevalence and incidence of depression²¹. However, it is not clear whether the association between neuroticism and depression among older people changes across subsequent birth cohorts. Previous studies in younger samples have reported higher neuroticism in later born cohorts²², while studies in older samples²³ report no birth cohort differences. In relation to the life-course perspective, it has been reported that there is a decrease in neuroticism with age during adulthood²⁴, while there are no age-related changes in neuroticism from mid- to late life²⁵.

The study of time trends in depression involves a number of methodological challenges. Studies should include representative population-based samples undergoing personal examinations, as register data (such as health care registers or prescription data) may be influenced by changes in awareness of depression among clinicians and patients. In addition, register studies only captures those seeking help at health care facilities, and only a minority of persons with depression are detected by the health care system. Examinations also need to be identical over time. These criteria are fulfilled in the Gothenburg H70 Birth Cohort Studies (the H70 study), which have been conducted over more than four decades.

Studies on time trends in depression can provide a dynamic view of population mental health over time. It can thus generate hypotheses for further research regarding the role of underlying factors for depression in the older population (e.g. regarding environmental factors or gender roles), which may have importance for creation of preventive programs and means of treating depression, but also be utilized in planning of psychiatric healthcare, need assessments, service planning and policy development. It can also give clues regarding the ability of the health care system to detect and treat depression, and whether this has changed over time.

We have previously reported that several factors related to depression in older age have changed during the last decades, such as improvement in cognitive function²⁶, lung function²⁷, cardiovascular health²⁸, sexual activity²⁹, and decrease in the prevalence of dementia^{30,31}. We therefore hypothesized that the burden of depressive symptoms and the prevalence of depression would be lower for both sexes in later-born cohorts. We further hypothesized that this change would be particularly pronounced in women due to the reported increase in gender equality³². Due to the strong association with depression, a lower level of neuroticism could also be expected.

Aims of the study

The first aim of this study was to explore birth cohort differences in burden of depressive symptoms, and prevalence of depression in representative samples of Swedish 70-year-olds examined with identical methods in 1976-77, 1992-93 (only women), 2000-02 and 2014-16, and whether time trends differed between sexes. Second, to further strengthen the evaluation of birth cohort differences in depression, we also examined time trends in neuroticism, and whether its relation to depression changed over time.

METHODS AND MATERIALS

Participants

The participants originate from the Gothenburg H70 Birth Cohort Studies in Sweden (the H70 studies)³³. Eligible participants were systematically sampled from the Swedish Population Register based on birth dates, and included persons living in both private households and residential care. In this study, examinations of 70-year-olds conducted in 1976-77 (birth cohort 1906-07), 1992-93 (birth cohort 1922, women only), 2000-02 (birth cohort 1930) and 2014-16 (birth cohort 1944) were included.

Examination 1976-77 (birth cohort 1906-07): In 1976-77, all 70-year-olds living in Gothenburg and born between July 1st, 1906 and June 30th 1907 on birth dates ending with 2, 5 or 8 were invited to participate (n=1281). All participants were numbered from 1 to 5. Those with number 1 or 2 were invited to take part in a psychiatric examination (n=513). Out of these, 404 (177 men and 227 women) participated (response rate 78.8%)³⁴. In this paper, participants for whom depression diagnosis could not be established due to missing data (n=5) and participants with dementia (n=7) were excluded, leaving 174 men and 218 women for analyses (n=392).

Examination 1992-93 (birth cohort 1922): In 1992-93, all 70-year-old women living in Gothenburg and born during 1922 on birth dates 6, 12, 18, 24 or 30 were invited to participate (n=473). A total of 299 participated (response rate 63.2%)³⁵. Out of these, 236 women took part in the psychiatric examination. In this paper, participants for whom depression diagnosis could not be established due to missing data (n=2) and participants with dementia (n=8) were

excluded, leaving 226 women for analyses.

Examination 2000-02 (birth cohort 1930): In 2000-02, all 70-year-olds living in Gothenburg and born during 1930 on birth dates: 3, 6, 12, 18, 21, 24, or 30 were invited to participate (n=753). A total of 524 participated (response rate 70%). Out of these, 499 (229 men and 270 women) took part in the psychiatric examination. In this paper, participants for whom depression diagnosis could not be established due to missing data (n=4) and participants with dementia (n=8) were excluded, leaving 227 men and 260 women for analyses (n=487).

Examination 2014-16 (birth cohort 1944): In 2014-16, all 70-year-olds living in Gothenburg and born during 1944 on birth dates ending with 0, 2, 5 or 8 were invited to participate (n=1667). A total of 1203 participated (response rate 72.2%)³⁶. Out of these, 1194 (555 men and 639 women) took part in a psychiatric examination. In this paper, participants for whom depression diagnosis could not be established due to missing data (n=9) and participants with dementia (n=19) were excluded, leaving 542 men and 624 women for analyses (n=1166).

Study Procedures

The examinations were conducted at an outpatient clinic or in the participant's home, and comprised comprehensive social, somatic, cognitive, functional, and psychiatric examinations, as well as close informant interviews and a battery of laboratory tests. All examinations were performed by psychiatric health professionals using identical methods over four decades to enhance possibilities of birth cohort comparisons.

The H70 studies were approved by the Ethics Committee for Medical Research at the

University of Gothenburg 1976-2002, and by the Regional Ethical Review Board in 2014. Informed consent was obtained from all participants or their close relatives, and the study was conducted according to the Helsinki Declaration.

Psychiatric examination

The psychiatric examination comprised a semi-structured interview conducted by psychiatrists in 1976-77 and 1992-93, by psychiatric research nurses in 2000-02, and by psychiatric research nurses, psychiatrists or medical doctors in 2014-16. The psychiatric nurses and medical doctors who performed the examinations from 1992-93 to 2014-16 were trained by a psychiatrist (IS) who, in turn, was trained by the psychiatrists who performed the psychiatric examinations in 1976-77. Interrater reliability³⁷ was assessed in 50 individuals who had dual ratings by either psychiatric nurses or psychiatrists. K values for the presence versus absence of symptoms and signs necessary to diagnose depression were between 0.62 and 1.00 indicating “good” (reference range K = 0.61-0.80) or “excellent” (K = 0.81-1.00) agreement¹⁶.

The psychiatric examination comprised ratings of psychiatric symptoms and signs experienced in the past month according to the Comprehensive Psychopathological Rating Scale (CPRS)³⁸, which has good reliability among older persons³⁹. Depressive symptom burden was rated according to the Montgomery-Åsberg Depression Rating Scale (MADRS)⁴⁰. MADRS is a subscale of the CPRS and include 10 items representing depressive symptoms. Individual items were rated from 0 (no symptoms) to 6 (severe symptoms), generating a MADRS-score ranging between 0-60. MADRS has been found to be valid among older populations⁴¹.

Neuroticism

The Eysenck Personality Inventory (EPI) comprises 57 self-rating questions (yes/no) and is designed to measure the personality dimensions extraversion–introversion and neuroticism–stability⁴². Neuroticism score ranges between 0–24. High scores represent emotional overreaction combined with low ego-strength, guilt proneness, anxiety and psychosomatic concerns. EPI data was not collected in 2000–02 (birth cohort 1930).

Diagnoses

Depression

Depression diagnoses were established using computerized symptom algorithms based on the CPRS, retrospectively applied to the responses from the psychiatric interviews at all examinations (1976–77, 1992–93, 2000–02 and 2014–16), in accordance with previous analyses from the H70 studies⁴³.

Major depression was diagnosed according to the Diagnostic and Statistical Manual of Mental Disorders Fifth Edition (DSM-5)⁴⁴, and required the presence of at least 5 out of 9 pre-specified depressive symptom clusters during the last month, of which at least one had to be depressed mood or diminished interest/pleasure. Minor depression was diagnosed according to DSM-IV-TR research criteria⁴⁵ and required the presence of 2–4 of the same pre-specified symptoms as for major depression. For the purpose of this paper, the term “any depression” was used to denote those fulfilling criteria for either major or minor depression.

Dementia

The diagnosis of dementia was used for exclusion in the main analyses, due to the difficulty of diagnosing depression when dementia is present. We were not able to diagnose

dementia according to DSM criteria in 1976–77. To enable comparisons between the four examinations, dementia was instead diagnosed according to criteria described by Kay et al ⁴⁶. These criteria were widely used in the 1970s and required the presence of severe disorientation in time or place or severe memory impairment as assessed during the psychiatric examination. In 2000-02, we were able to diagnose dementia according to both the old criteria and DSM-III-R ⁴⁷. The observed agreement between the two classification systems was high ($K = 0.81$) ⁴⁸.

Demographic factors

All demographic factors are based on self-reported information. Educational level was defined as ‘compulsory’ (i.e. ≤ 6 years in those born 1906-07 and 1922, ≤ 7 years in those born 1930, and ≤ 9 years in those born 1944) vs. ‘more than compulsory’. Regulations for mandatory years in Swedish primary schooling have changed several times during the 20th century, leading to different cut-off points for educational level in this study. Type of residence was categorized as ‘sheltered accommodation’ vs. ‘private household’. We also asked if the participant was living alone or not. ‘Having partner’ included married, having non-cohabiting partner, and having cohabiting partner. We also asked if the participant had a happy relationship, if they had children, if they felt lonely, and if they had lost their partner during last 5 years due to death or divorce. All participants were asked to report their medications, doses and indications for treatment. Antidepressants (N06A) were classified according to the Anatomical Therapeutic Chemical (ATC) classification system recommended by the WHO ⁴⁹. We also asked if the participant was experiencing back or joint pain (joint pain also included stiff or swollen joints). Having contact with health care included contact with medical doctor or nurse at any health care facility or during home visit during the last 3 or 12 months.

Statistical Analysis

Pearson's Chi-square was used to test differences in proportions. To compare differences in group means in MADRS and neuroticism score, we utilized the Analysis of Variance (ANOVA) statistical model with the flexibility to perform post-hoc Tukey test. Independent samples t-tests were used to test sex differences in mean scores. A univariate Generalized Linear Model (GLM) was used to check for potential effect modification by sex on birth cohort differences in mean MADRS score and mean neuroticism score, where the interaction term sex*birth cohort was added, and for potential effect modification by neuroticism on sex differences in depression, where the interaction term neuroticism*sex was added. A binary logistic regression (crude odds ratios with 95 % confidence intervals) was used to test differences by birth cohort and sex in the prevalence of major, minor and any depression, and to check for potential effect modification by sex on birth cohort differences in depression, where the interaction term sex*birth cohort, was added.

Binary logistic regression was used to test for possible associations between mean neuroticism score and major, minor, and any depression. A linear regression model was used to test for possible association between mean neuroticism score and mean MADRS score.

Sensitivity analyses using log-transformed values ($\log_{10}$) for MADRS were performed, supporting the results by reducing the effects of outlying observations in the MADRS score. Sensitivity analyses not using dementia as exclusion criteria were performed for all birth cohort comparisons in depression and neuroticism, supporting the results. Statistical methods were carried out using IBM SPSS STATISTICS 22. All statistical tests were two-tailed and p-values of <0.05 were considered statistically significant.

RESULTS

Sample characteristics of 70-year-olds examined 1976-77, 1992-93, 2000-02 and 2014-16 are presented in Table 1. As may be seen, the use of antidepressants was higher among later born cohorts.

(Insert Table 1)

Burden of depressive symptoms

Data on depression in 1992-93 were available only for women.

Birth cohort differences in MADRS score

As may be seen in Table 2, MADRS score was lower in 2014-16 compared to 1976-77 in the total sample. Among women, MADRS score was lower in 2014-16 compared to 1992-93 and 1976-77, but did not differ from 2000-02. No significant time trends were observed among men. Among women without depression, MADRS score was lower in 2014-16 compared to 1976-77, but was not significantly different from the other examinations. No differences were observed among men, or among those with depression.

Interaction effect

There was an interaction between sex and birth cohort in relation to MADRS score ($B = -0.46$; $R^2 = 0.02$; $p = 0.02$), i.e. sex differences declined, and the decrease in MADRS score at later examinations was more accentuated in women.

(Insert Table 2)

One-month prevalence of depression

Figure 1 shows the one-month prevalence of major, minor and any depression in 70-year-olds from 1976-77 to 2014-16.

Birth cohort differences in depression diagnoses

The prevalence of major depression did not differ between examinations in the total sample. The prevalence of minor depression and any depression was lower in 2014-16 compared to 2000-02. Among women, the prevalence of any depression was lower in 2014-16 compared to 2000-02 and 1992-93. The prevalence of major depression in women was lower in 2014-16 compared to 1992-93. The highest prevalence in women was observed in 1992-93. No birth cohort differences were found among men.

Interaction effect

There was no interaction effect between sex and birth cohort in relation to minor depression ($B = -0.13$; $OR = 0.88$; $p = 0.39$), major depression ($B = -0.06$; $OR = 0.94$; $p = 0.82$) or any depression ($B = -0.12$; $OR = 0.89$; $p = 0.36$).

(Insert Figure 1)

Neuroticism

Data on neuroticism were available only for women in 1992-1993, and data collection was not conducted in 2002-02. For 1976-77, 1992-93, and 2014-16, independent samples t-tests showed small or no difference in mean MADRS score between those having missing data on neuroticism and those who did not.

Birth cohort differences in neuroticism

Mean neuroticism score was lower in 2014-16 compared to 1976-77 (6.1 vs. 7.8; $p<0.01$) in the total sample. As may be seen in Table 2, neuroticism in women was lower in 2014-16 compared to both 1992-93 and 1976-77, while no significant time trends were observed among men. Among those without depression, neuroticism decreased both among women and among men. Among women with depression, neuroticism was lower in 2014-16 compared to 1976-77.

Association between neuroticism and depression across birth cohorts

There was an association between neuroticism and depression at all examinations. Overall, this association did not change across cohorts. The results can be seen in Supplementary Table 1.

Interaction effect

There was an interaction between sex and birth cohort ($B = -0.64$; $R^2 = 0.07$; $p < 0.01$), i.e. the association between sex and neuroticism score differed between examinations (the sex difference declined), and the lower neuroticism score at later examinations differed between sexes (decline only observed in women). There was a tendency towards an interaction effect between sex and neuroticism in relation to major depression in 1976-77 ($OR = 1.4$; $p = 0.07$), not reaching our set level of statistical significance, while no interaction was found in 2014-16 ($OR = 0.9$; $p = 0.57$). There was no interaction in relation to minor depression, neither in 1976-77 ($OR = 0.9$; $p = 0.54$), nor in 2014-16 ($OR = 1.0$; $p = 0.91$). There was no interaction in relation to

any depression neither in 1976-77 (OR=1.1; $p=0.41$), nor 2014-16 (OR=0.9; $p=0.93$). There was an interaction effect between sex and neuroticism in relation to MADRS score in 1976-77 ($R^2=0.25$; $B=0.36$; $SE=0.13$; $p=0.006$), while there was no such interaction in 2014-16 ($R^2=0.27$; $B=0.06$; $SE=0.07$; $p=0.34$).

Sex differences

Table 3 shows sex differences in MADRS score, the prevalence of depression, and neuroticism, by examination year. Women had higher MADRS score, higher prevalence of any depression, and higher neuroticism score compared to men at all examinations. For any depression, the sex ratio was 2.6:1 in 1976-77 ($p<0.05$), 2.0:1 in 2000-02 ($p<0.05$), and 1.7:1 in 2014-16 ($p<0.05$). While no significant sex differences were observed for major and minor depression (except for minor depression in 1976-77), the female:male sex ratio ranged between 2.7:1 to 1.5:1.

(Insert Table 3)

The use of antidepressants

Despite an increased use of antidepressants between 1976-77 and 2014-16 (4.5% vs. 9.7%; $p<0.01$) in the total sample, only a minority of persons with depression received pharmacological treatment (18.2% in 1976-77 vs. 27.7% in 2000-02 vs. 30.8% in 2014-16; $p>0.05$). The use of antidepressants among women having depression was higher in 2014-16 than in 1976-77 (39.7% vs. 18.2%; $p<0.05$), while no differences were observed among men. Birth cohort differences in antidepressant use among persons not having depression were observed in both men (1.2% in 1976-77 vs. 5.3% in 2014-16; $p<0.05$) and women (4.2% in

1976-77 and 4.9% in 1992-93 vs. 9.8% in 2014-16; $p<0.05$). Antidepressant use was more common in women compared to men at all examinations (6.3% vs. 2.3%, 13.5% vs. 3.1%, 13.1% vs. 5.9%; $p<0.05$).

Post-print

DISCUSSION

We found that depressive symptom burden and neuroticism decreased among women but not among men between 1976 and 2016 among a population-based sample of 70-year-olds. Time trends for the diagnoses of major, minor, and any depression were less clear. The lowest prevalence for minor depression was noted in 2014-16, while that of major depression remained stable. Women still had higher burden of depressive symptoms, neuroticism and prevalence of any depression compared to men at all examinations, a global phenomenon in all age groups^{2,50}. However, the gender gap decreased over time.

The prevalence of depression may be influenced by how life courses are embedded in societal and geographical contexts across historical times⁵¹, i.e. by age, birth cohort, and period effects. Disentangling birth cohort and period effects can be challenging. Birth cohort effects may have generated differences in health-related factors throughout the life course, which may be related to the proneness of depression. For example, our cohort born 1906-07 experienced starvation during World War I, food rationing during World War II, had poor living conditions during the first 3-4 decades of their lives (including poor education, working conditions and health care), while the latest born cohort of 1944 lived their lives in a modern welfare state. A period effect may be that we had the highest depression rate in 1992-93, when an economic recession took place in Sweden. As only women were examined in 1992-93, we do not know if the peak was present also for men (see Figure 2).

(Insert Figure 2)

For any depression, the sex ratio decreased from 2.6 in 1976-77, to 2.0 in 2000-02, and 1.7 in 2014-16. One possible reason for this finding may be the increasing gender equality

occurring in Sweden during the 20th century⁵². Later born cohorts of women have to a larger extent benefitted from societal improvements, such as women's emancipation during the 20th century, the large expansion of governmentally funded child day care in the 1960s and 70s, the sexual revolution in the 1960s with the introduction of contraceptive pills, free abortion in the 1970s, increased quality of pre- and postnatal care, increased standard and accessibility of health care, and access to university education and paid employment. Still, due to cultural gender norms, women face a role strain overload⁵³, expected to manage their own work, primary household, care for children, grandchildren and spouses, even beyond retirement age. This overload has been associated with depressive symptoms in women⁵⁴, and might partly explain why the prevalence of depression and burden of symptoms still remained higher among women compared to men across our 40 year study period, in spite of societal improvements. Another factor that can be related to sex differences in depression is rumination. This is a type of repetitive negative thinking, which is reported to be both more common in women and also a risk factor for depression⁵⁵. Other suggested explanations for the higher prevalence of depression in women includes that women report symptoms more often while men do not perceive or report them⁵⁶, or the presence of biological differences, e.g. related to hormonal⁵⁷ or genetic factors⁵⁸.

Our findings highlight the importance of environmental factors. Previous studies²² have shown that environmental factors (e.g. rates of unemployment, economic recession, or worries about climate change or terror attacks) play a large role in birth cohort differences in personality and anxiety, which are both strongly associated with depressive mood. The direction of change in environmental factors can merely be speculative, causing future time trend projections for depression in late life to be uncertain. A worrying trend is that the prevalence of depression is increasing in younger age groups¹³, which might have implications for future generations of

older people due to the potential risk of recurrence ^{8,59}. However, studies also show that emotional wellbeing improves with age ⁶⁰.

Some of the environmental factors displayed in Figure 2 may have affected both sexes differently (e.g. rationings during WWI and WWII, the recessions in the 1930s and 1990s, access to improved health care, or development of treatment methods for depression). As an example, a previous study showed that increased rates of unemployment due to economic recessions may affect the time trends of recorded depressions (at general practices) among men, but not among women ¹². A continuing of societal improvements (e.g. increased awareness of depression that reduce stigma, changes in normative masculine gender roles allowing men to recognize and express depressive symptoms, enhanced treatment efficacy, increased equality in legal rights and societal position for men and women) and improved life conditions (e.g. life expectancy, burden of disease), may generate a further reduction in the sex ratio of depression. The reason why we do not find any time trends in the prevalence of depression among men may not primarily be due to less significant benefits from societal improvements (compared to the benefits that women may have gained), but also partly due to small sample sizes, or consolidated masculine gender roles still connected to the under-reporting of depressive symptoms. However, the higher prevalence of depression among women (compared to men) is so far a robust finding irrespective of time period, age of samples, and geographical settings.

Results on time trends and birth cohort differences of depression may vary between studies depending on examination years, time spans, birth cohorts, age ranges, and geographical areas. Also, contradicting results between studies can be due to the use of different instruments and diagnostic criteria, which previously has been shown in a Swedish study of older adults ⁶¹. The only change between DSM-III/DSM-IV and DSM-5 is that bereavement is no longer an exclusion criterium for a diagnosis of depression. However, in our study, diagnoses were made

(using retrospectively applied symptom-based algorithms) according to DSM-5 (major depression) and DSM-IV-TR research criteria (minor depression) at all examinations (1976-77, 1992-93, 2000-02 and 2014-16) in order to increase comparability.

Few studies on time trends in depression among older persons have so far extended beyond 2010. Several recent studies extending into the 2000s and 2010s report declining frequency of minor or any depression among older persons^{5,12}. A Dutch study on 55-65 year-olds reported a decrease in past-year subthreshold depression (comparable to minor depression in our study) between 2002 and 2012 for both sexes combined, but an increase in major depression from 1992 and 2002¹⁴. Studies on major depression among older persons conducted during the same period report stable prevalence^{15,16,62}. In contrast, time trend studies focusing mainly on younger and middle-aged populations generally report higher prevalence of depression and depressive symptoms among later born cohorts^{13,14,17,54,63-66}, but the prevalence is also reported to be stable over time^{67,68}.

We have previously reported an increase in the prevalence of minor depression among 75-year-olds between 1976-77 and 2005-06¹⁶. Reasons for this contrasting finding may be the older age (75 vs. 70), and that the cohort born 1930 was compared with a cohort born 1901-02. In our present study, the lowest prevalence was observed in the cohort born 1944. We could not confirm previous findings suggesting that persons born after World War II have higher prevalence of depression than those born earlier in the 20th century (age >18 years)⁶⁹⁻⁷¹. One reason could be that these studies were performed in the US during the 1980s and 90s, when our cohort born 1944 was only about 40 years old. It should be noted that frequency of depression may fluctuate over time and between birth cohorts, as described in a study from US and Canada in 1997-2010¹⁷. Few other studies have a time span of more than four decades

^{68,72,73}, and these only extends into the 1990s. Among studies with large time spans, the Stirling County Study in Canada, conducted 1952-1992, reported a stable prevalence of depression over time ⁶⁸, and the Lundby Study conducted 1947-97 reported a decreasing incidence of depression in 1972-1997 compared to 1947-1972 ⁷³.

The life expectancy has increased between 1976 (approximately 72 for men and 78 for women) and 2014 (approximately 80 for men and 84 for women) ⁷⁴. Depending on the general health status of the aging population, increased life expectancy may be associated with both higher and lower prevalence of depression. A higher life expectancy may increase the prevalence of depression later in life as more people with depression survive to older ages. On the other hand, a higher life-expectancy may be a reflection of better health, which may lead to lower prevalence of depression. In addition, if the relative life-expectancy increases more in people without depression than in those with depression, the prevalence will also decrease. Previous findings derived from the Gothenburg H70 Birth Cohort Study show improved health status for successive birth cohorts ²⁶⁻³⁰. Therefore, the increased life expectancy and improved life conditions in Sweden may partly be a reason for the declining prevalence of depression among 70-year-olds.

We also noted a decline in neuroticism measured with a self-rating scale between 1976-77 and 2014-16. It has been suggested that neuroticism and depression share etiological factors ^{75,76}, and may therefore be affected by similar birth cohort effects. In support of this, we found a strong association between neuroticism and depression at all examinations, in line with previous studies ^{19,77}. This association did not differ by birth cohort. Our finding of lower neuroticism score in later born birth cohorts support the finding of a lower prevalence of depression in 2014-16, as data collection methods differ (self-report questionnaire vs.

psychiatric interview). However, it can be discussed whether the lower score of neuroticism is the reason for the lower burden of depressive symptoms, or vice versa. The direction of this association cannot be determined from this repeated cross-sectional study. However, we have previously reported that neuroticism predicts later episodes of depression among older persons²¹. It could be noted that neuroticism also decreased among persons without depression.

Although not the main focus of this paper, we also analyzed birth cohort differences in the use of antidepressants to have the full picture of depression in the population. The introduction of second generation antidepressants constitute a possibly important period or birth cohort effect, although we acknowledge that our study cannot estimate to what extent this has causally influenced the prevalence of depression. As expected, we found increased use of antidepressants after the 1990s, with a stable prevalence after 2000, in line with a Canadian study 1994-2012⁷⁸. Paralleling previous studies, we found that the majority of persons with depression were not using antidepressants^{6,67,79}. However, treatment rate for depression increased substantially between 1976-77 and 2014-16 (from 18% to 40%) in women with depression, but not in men. Increased treatment rate may indicate an increased detection rate of depression in women in the health care system, or that physicians are prescribing for broader indications among women than previously. The sex difference may also be due to non-compliance among men, that men have other treatments for depression, or that they had not yet sought help within the health care system. We cannot elucidate if antidepressants have been prescribed for treating depression, if its increased use is the reason for the lower prevalence of depression, or if treatment with newer antidepressants may be more or less effective. Also, there is a possibility that some persons on antidepressants at the time of the examination have previously fulfilled criteria for depression and have been successfully treated. It has been suggested that increasing rates of antidepressant treatment (since the introduction of SSRI in

the 1990s) has led to a decrease in suicide rates in Sweden⁸⁰. This is, however, debated⁸¹, and may not have influenced suicide rates among older persons.

Strengths and limitations

Among strengths in this study are the comprehensive personal examinations of representative samples using identical methods over 40 years, that diagnoses of depression are based on psychiatric examinations conducted by psychiatrists and psychiatric nurses using structured interviews, and the use of computerized algorithms to diagnose depression, which minimizes effects of altered routines for clinical evaluations and shifting diagnostic boundaries. Also, this procedure is less influenced by potential changes in awareness of depression and help-seeking behaviors compared to information collected from registries or health records. In addition, our diagnoses are based on past-month symptoms, which reduces recall bias⁸².

There are also some limitations. First, the study was conducted in a mainly Caucasian Swedish sample, and our results are foremost compared with studies conducted in high income countries, since birth cohort comparisons of depression among older populations are rare in other parts of the world. We can therefore not generalize to other populations. Second, despite fairly high response rates at each examination, we cannot exclude the possibility of different time trends among participants and non-responders. Third, the investigating staff has altered between psychiatrists, psychiatric nurses, and medical doctors over the years, which to some extent could add to the differences in depression prevalence. However, all investigating staff was trained by the last author (IS), who in his turn was trained by the psychiatrists who performed the examinations in the 1970s. The high kappa value for interrater reliability indicates consistency in examinations over time. Also, neuroticism was measured using a self-rating questionnaire, while depressive symptoms were assessed during a psychiatric interview.

550 This adds credibility to the fact that the lower burden of depressive symptoms and prevalence
551 of any depression is not due to measurement inconsistency between staff at the different
552 examinations. Fourth, the 1922 birth cohort only includes women, which limited our ability to
553 investigate time trends in men to the same extent. Fifth, some sub-groups in our analyses (e.g
554 men and women with major depression) were small, which limited the statistical power and
555 may have generated false negative results, e.g. regarding sex differences in major depression.
556 Sixth, comparable data on physical diseases was not available for the cohorts included in this
557 paper. This limited our ability to analyze whether possible birth cohort differences in physical
558 diseases may be a reason for the lower prevalence of depression in later-born cohorts. However,
559 we take note of the fact that we have previously reported that several factors related to
560 depression in older age have changed during the last decades, such as improvement of lung
561 function ²⁷, and cardiovascular health ²⁸ in 75-year olds born 1901-02 and 1930, improvement
562 in cognitive function ²⁶ and sexual activities ²⁹ among 70-year-olds born 1901-02 and 1930, and
563 decrease in the prevalence of dementia ³⁰ between 85-year-olds born 1901-02 and 1923-24.
564 Seventh, due to the difficulties in establishing reliable depression diagnosis in persons with
565 dementia, dementia cases were excluded for all main analyses in this paper. Thus, the
566 prevalence of depression among Swedish 70-year-olds may partly be underestimated. However,
567 the prevalence of dementia in our samples only ranged between 0.9-3.4 %, and the results from
568 the sensitivity analyses including dementia cases showed that this did not affect the results.

569 Finally, stigma for depression has decreased between the 1970s and 2010s. This might
570 have underestimated the prevalence of depression in the oldest birth cohort.
571

572 In conclusion, our results show that depressive burden and neuroticism decreased in
573 women between 1976 and 2016. The decreased sex ratio in depression and neuroticism in later-
574 born birth cohorts of 70-year-olds may be due to the increasing gender equality occurring in

575 Sweden during the period and shows the importance of considering environmental factors in
576 the discussion on sex differences in depression.

Post-print

Dealing with Shame: Saudi Mothers' Responses to Their Children's Diagnosis of Autism Spectrum Disorder

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Abstract: Autism is a disorder affecting individuals' academic, communication, and social development. The findings of this qualitative study report 15 Saudi mothers' responses to their children's diagnosis of autism spectrum disorder and provide insights into mothers' experiences and responses to stigma in the context of the Kingdom of Saudi Arabia.

Keywords: Autism Spectrum Disorder; Stigma; Mothers' Perceptions

Introduction

Individuals diagnosed with autism spectrum disorder (ASD) have difficulties when communicating and often show repetitive interests and behaviors. Therefore, mothers of children with this condition have encountered many problems when raising them. How ASD manifests depends on the chronological age and developmental level of the children. Symptoms of ASD include delayed toilet training, temper tantrums, self-injurious behavior, aggression, impulsivity, and hyperactivity (Duarte, Bordin, Yazigi, & Mooney, 2005). On the other hand, mothers of children diagnosed with ASD are more likely to experience depression, higher pessimism about their children's future, and reduced quality of life (Cappe, Wolff, Bobet, & Adrien, 2011; Johnson, Frenn, Feetham, & Simpson, 2011). According to previous research, it is evident that parents of infants diagnosed with ASD experience more parenting stress relative to other individuals (Hayes & Watson, 2013), indicating that having children with ASD can strongly impact the health and well-being of parents. Consequently, the degree of parental stress is proportional to the severity of children's behaviors and symptoms of autism (Davis & Carter, 2008; Ingersoll & Hambrick, 2011; Lyons, Leon, Phelps, & Dunleavy, 2010). When children manifest poor behavior and severe symptoms, their parents subsequently experience higher parental stress.

Furthermore, parents of children diagnosed with ASD often feel shame. For example, they may experience disapproval and criticism from neighbors as they are perceived as being unable to control their children (Gray, 1993; Mak & Kwok, 2010). Due to this stigma,

children diagnosed with ASD are perceived as behaving inappropriately of their own volition. As a result, many individuals think that parents are to be blamed for using ineffective parenting styles when raising their children (Fernandez & Arcia, 2004). Consequently, mothers might feel they are not utilizing appropriate interventions or treatments, specifically when their friends advise them about various other treatments or interventions they should use (Miller, Schreck, Mulick, & Butter, 2012). Additionally, mothers may blame themselves for not being able to help their children develop properly (Kuhn & Carter, 2006). They might even blame themselves for being the reason for their children's diagnosis. Therefore, these experiences of self-blame, shame, and guilt are the major causes of anxiety, depression, and parenting stress (Cappe et al., 2011; Mak & Kwok, 2010).

The shame, blame, and stigma experienced by individuals with disabilities and their families are reflected in the moral and medical models of disability where fault is attributed to the individual or his/her family and not to society (Olkin, 2002). Olkin (2002) proposes in the moral model of disability, that disability "is a defect caused by moral lapse of sins, failure of faith, evil, test of faith...brings shame to the person with the disability and his or her family" (p. 133). The moral model results in individuals "being ostracized from family and community, having profound shame, having to hide disability symptoms or the person with a disability" (Olkin, 2007, p. 6).

Review of the Literature

This study examined mothers' responses to the cultural stigma associated with their children's diagnosis of autism and the role institutions play. Individual interviews with mothers allowed for an examination of these women's experiences navigating issues of stigma related to autism in the Kingdom of Saudi Arabia (KSA). Goffman's (1963) theory of stigma as a theoretical framework underpinning this research review and stigma is examined in relation to culture, paying particular attention to the role that the Saudi culture plays in the lives of mothers whose children have been diagnosed with autism.

Culture

Culture plays a role in shaping the way individuals see, think, and understand the world. It comprises "a dynamic yet stable set of goals, beliefs, and attitudes shared by a group of people" (Gurung, 2006, p. 172) and even though some beliefs are fluid, "the general level of culture stays stable because individuals change together" (Gurung, 2006, p. 4).

As members of a community, the lives of individuals with ASD are generally determined by the culture in which they are immersed. Families and communities do the best they can to help individuals with ASD function as productive members of society and live as normally as possible, guided by their cultural beliefs and level of understanding. The beliefs held by members of a community regarding the potential and treatment of individuals with autism differ from one culture to another and can change over time (Ravindran & Myers, 2012). Culture influences key aspects of the diagnosis and treatment process, and affects the likelihood that individuals seek help, what treatments to use, available resources, and the

relationships between families and professionals. Daley (2002) suggests treatment approaches that work and are culturally accepted in the West may not be the same approaches accepted and used to treat individuals with ASD in other parts of the world. For instance, in the West, known causes of autism are increasingly attributed to science and various treatments that are available which include behavioral, cognitive, pharmaceutical, sensory, relational, vitamins, and diet therapy. However, in other cultures, the causes of autism might be regarded as “‘Karma,’ ‘Allah’s will,’ or ‘the will of God’” and may be treated with non-scientific remedies such as acupuncture and/or herbal medicine (Ravindran & Myers, 2012, p. 311).

ASD and Stigma

Within the scientific community, autism is viewed as a disorder marked by abnormal social skills stemming from an incurable medical condition (Martin, 2012). Over the years, the negative stigmatization and social perceptions associated with ASD have grown due to the individualized nature of the syndrome and a lack of understanding of the symptoms (Martin, 2012). Historically, one of the most common social challenges faced by individuals with autism is a feeling of isolation as they grow older, due to increased segregation and rejection from peers, combined with an increase in self-awareness (Martin, 2012). An individual’s level of isolation is dependent on key factors such as environment, personality, and intensity of impairment, all being impacted by the level of stigmatization received from others in the surrounding environment. Isolation and rejection are further exacerbated by the fact that the social behaviors of individuals with autism are perceived as different from the norm, or deviant. As such, since the behaviors of those with autism do not meet the expectations of what is appropriate, they are often ignored and avoided by the normative groups.

Goffman’s Theory of Stigma

Goffman’s (1963) framework is useful for examining Saudi mother’s perceptions of their experiences with stigma associated with their children’s autism. Particularly when discussing the societal stigma associated with having certain characteristics, Goffman sheds light on how communities stigmatize individuals that display certain traits that are deemed undesirable. This is true for individuals with autism in Saudi Arabia, where the culture still stigmatizes such individuals.

In *Stigma: Notes on the Management of Spoiled Identity*, Goffman (1963) describes the different types of stigma and how they impact the identity of those who are labeled undesirable by society. Goffman refers to those who are not forced to live under a perceived stigma, or “do not depart negatively from the particular expressions at issue” as “normals” (p. 5). This perspective is in line with how society views individuals who have been stigmatized, as well as, behaving the opposite of how normal society should act, despite how damaging these thought processes are to the relationships between the stigmatized and the general population.

Goffman (1963) identifies three separate forms of social stigma that are essential to conceptualizing how stigma impacts members of a society, whether individuals are, or may be perceived through this ever-present lens. The first form of social stigma involves physical abnormalities that are referred to with harsh, unsympathetic words. Goffman writes these

“abominations of the body—the various physical deformities” are enveloped with harsh judgment and are prevalent despite the physical differences occurring without the person’s will or desire (p. 4).

The second form of social stigma is associated with characteristics or behavioral traits that bring negative connotations despite often being no direct connection between the trait and the negative association. Often, these traits are:

Blemishes of individual character perceived as weak will, domineering, or unnatural passions, treacherous and rigid beliefs, and dishonesty, these being inferred from a known record of, for example, mental disorder, imprisonment, addiction, alcoholism, homosexuality, unemployment, suicidal attempts, and radical political behavior (Goffman, 1963, p. 5).

As seen in this quotation and thousands of other representations of people who express certain traits associated with the stigma, each action or trait can trigger a far worse reaction and behavior in turn.

The last form of social stigma is tribal stigma which is associated with linking different identities and people of a larger group into a singular category. It is created by combining individuals of a specific race, religion, nation, or some form of community into one single group or person, rather than a group of unique individuals. The individualistic needs and components of each person in the stigmatized community are immediately erased and those in the broader outside community are unable to sympathize with those stigmatized. Outsiders, on the other hand, relieve themselves from any responsibility by disassociating themselves from those in the stigmatized community. Therefore, dehumanizing and apathetic responses towards the community become easier and much more frequent (Goffman, 1963).

Goffman (1963) reinforces this statement by continuing to describe what he calls the stigma theory which allows the community to continuously dehumanize someone with any stigmas and justifies whatever negative issues/perceptions in regard to the stigmatized individuals, as well as, continues to “break the claim that his other attributes have on us” (p. 5). Goffman notes that any other claim or connection that the stigmatized persons have with the general population is put aside in comparison to the overriding stigma, whether from physical, behavioral, or group stereotypes of the stigma.

In the case of autism, social stigma arises from misconceptions, stereotypes, and ignorance of the disorder and those living with it. One can attribute this stigma towards individuals with autism to the fact that research on how to best diagnose and work with autism disorder has only begun to surface relatively recently. Despite research addressing stereotypes, stigmas have unfortunately continued.

Stigmas associated with autism are often based on the three types of stigmas as noted by Goffman (1963), with some stigmas related to how an individual with autism should look, but even more so, the perceived behavioral traits of someone with autism, and the assumed

identity grouping of the community of individuals with autism. Educators are often seen as abandoning the needs of children with autism by grouping them as a separate and distinct category from what Goffman would have referred to as normal children. In this way, some educators may excuse themselves from learning and helping children simply because of stereotypes based on the assumed behaviors that an individual with autism is expected to display. In response, the general population fails children by perpetuating the common stereotypes and resultant stigma in the classroom when a child with autism arrives.

More recently, Bos, Pryor, Reeder, and Stutterheim (2013) highlighted Goffman's contribution concerning stigma. Bos et al. (2013) and Olkin (2002) reiterate that stigma was originally conceived as a physical mark to identify immoral people, but over time, the definition incorporated behavior resulting in social disapproval. According to the authors, stigma varies across social contexts, and from a psychological perspective, stigma serves several functions, which include exploiting and domineering people, enforcing social norms, and avoiding diseases. Additionally, the authors discuss the theoretical views of the concept of stigma and further classify four additional types of stigma, which are public stigma, self-stigma, stigma by association, and structural stigma (Bos et al., 2013), and defined in the next section.

Bos et al. (2013) highlights the interrelationships between the types of stigma. First, public stigma arises from the cognitive representations that are perceived against people with a particularly stigmatized condition, especially if it is fatal or severely disabling. An example of public stigma is the assumptions and the representations a culture has for certain individuals. For example, different cultures have different representations for individuals with a physical impairment based on what they can and can't do. Self-stigma, on the other hand, results from the existence of public stigma, since people experiencing stigma understand the social devaluation attached to their condition. An individual who experiences stigma is likely to associate negative attitudes of others directed at them with their condition. Next, stigma by association affects individuals who have an association with stigmatized people. Mothers of individuals with stigma can experience this type of stigma as well as other relatives of the stigmatized individual. Finally, structural stigma presents the ways in which the society perpetuates a stigmatized condition. Society can perpetuate the feeling of stigma by continuing to associate labels with the condition.

Goffman's (1963) work is crucial for its ability to provide a framework through which stigma could be understood based on its characteristics and causes. Using this framework to understand stigma in a KSA context helps to make sense of mothers' experiences, by highlighting not only the cause of the stigma but how it operates within society.

The Saudi Context

According to Alqahtani (2012), most parents whose children have been diagnosed with ASD argue that cultural, medical, and psychological factors are the main causes of autism. The medical factors perceived to cause autism include vaccination, examination during pregnancy, and vitamin deficiency. Additionally, the cultural factors perceived to cause autism include black magic and the evil eye. Finally, perceived psychological factors

are due to the trauma of experiencing the death of parents and associated with causing autism.

Despite the existence of sufficient evidence showing the causes of autism spectrum disorder, parents blame themselves for their children's condition (Alqahtani, 2012). Since parents do not have adequate information about the causes of autism, they use interventions such as consulting religious leaders and reading Qur'an verses (Alqahtani, 2012). They prefer these treatments because they believe they heal the symptoms of autism. The findings of the study show that no parents used essential therapies focused on developmental, educational, and behavioral interventions (Alqahtani, 2012). Furthermore, it was found that some parents fear seeking help for their children due to the cultural stigma associated with disabilities (Alqahtani, 2012; Zeina, Al-Ayadhi, & Bashir, 2014).

Some individuals applaud women in the KSA for doing a commendable job as caregivers despite the lack of paying jobs (Basaffar, Niehm, & Bosselman, 2016). However, it is crucial to recognize that in Saudi Arabia, women and men play clearly distinct gender roles in the society. As a male-dominated community, men are perceived as the providers for the family while women perform traditional roles even when they are part of the labor force. Similarly, in homes where both parents are working, women are expected to meet the needs of their husbands and take care of their children. This means women have to balance their family and work life in order to be able to satisfy the needs of their children. The men may provide support and help, but they do not play active roles in overseeing the needs of children, including those who require special attention.

The findings are clear that parents in the KSA fear their children will be humiliated due to their autism diagnosis. Children with autism require special attention from parents or caregivers. Therefore, institutions have a responsibility and role to lessen the effects of social stigma that mothers of children with ASD experience. The findings of the current study bring to light the cultural issues associated with raising children with autism and the roles played by institutions in addressing issues of stigma.

Method

This study examined mothers' experiences with stigma associated with their children diagnosed with ASD in the KSA. Using Goffman's (1963) theory of stigma as a theoretical framework, 15 interviews were conducted with mothers in the KSA whose children were diagnosed with autism spectrum disorder. A phenomenological approach was used to explore these mothers' individual, personal experiences, examining the similarities while highlighting the uniqueness of their individual experiences. It allowed a better understanding of their experiences by providing access to their unique perspectives (Creswell, 2007; Marshall & Rossman, 1999; McCaslin & Scott, 2003; Wertz et al., 2011).

A phenomenological approach works from the belief that "there is an essence or essences to shared experience" (Patton, 2015, p. 116). Phenomenologists are interested in shedding light on what individuals experience and more importantly how they experience it.

Through its focus on individual experiences, a phenomenological approach works to validate these experiences. In addition, van Manen (1990) posits that a phenomenological approach should strive for collecting stories about the experiences of individuals and validating those experiences. While it allows for capturing the individual experiences and addressing their uniqueness, a phenomenological approach is described as “the study of the shared meaning of the experience of a phenomenon for several individuals” (McCaslin & Scott, 2003, p. 449). Due to a focus on shared experience, commonalities between the mothers were examined while remaining faithful to the uniqueness of their experiences.

The participants involved in this study included 15 mothers between 22 and 43 years old. They provided information about their experiences with stigma associated with raising children diagnosed with ASD. The study did not focus on the severity of the children’s condition or gender. It included all mothers with children between the ages of 5–12. The mothers provided unique responses to their experiences of stigma. Mash and Wolfe (2010) indicated parents’ reactions depend on their level of education, family setting, and work environment. Parents’ level of education may impact individuals’ reactions to the problems they face based on various forms of social, economic, and cultural stress (Mash & Wolfe, 2010).

Data Collection

Before conducting interviews, mothers involved in the study provided informed consent. The setting for each of the 25–45 minute individual interviews was a local coffee shop. They were audio recorded to ensure accurate capturing of participants’ responses. At the end of each interview, a summary of the interview was written, and transcriptions of interviews were created along with additional investigator memos.

Results

The investigation focused on the challenges associated with autism in a male-controlled society, specifically on mothers as they assume the role of primary caretaker for their children diagnosed with autism. Mothers’ responses were organized into themes/sub-themes for the following purposes. First, it represents the association of themes/subthemes. Second, it provides clear insights into mothers’ responses to the important issues that arose in the conversations. Third, it highlights the different issues that mothers chose to emphasize, struggled with, or found useful in their journey of trying to provide the proper services for their children. Finally, others reading this article are able to examine the responses on their own and evaluate the approach followed in analyzing the data.

For the ease of data presentation, a numbering system was used. M1 was used to represent the response of mother number 1 as found in the transcript. M1-1 was used to refer to a comment’s position in the conversation containing mother number 1. For example, M1-1 means that mother number 1 provided the comment as the first response in the interview transcript. The numbering system was applied for all the participants. This system allowed easy examination of the transcripts and verification of the conclusions drawn from mothers’

responses.

Mothers' Responses

The findings represent mothers' responses to their children's situation which are represented in Figure 1. First, the initial emotional responses mothers displayed in reaction to their children's condition are highlighted. Next, following their initial responses, mothers decided to take action to learn about their children's situations, and in addition to the positive steps they took, some relied on their faith to help them cope with the challenges they were facing. The findings conclude with an examination of mothers' perceptions of the negative effects of their children's autism on their social milieu.

Figure 1. Visual Representation of the Components of Mothers' Responses to Shame.

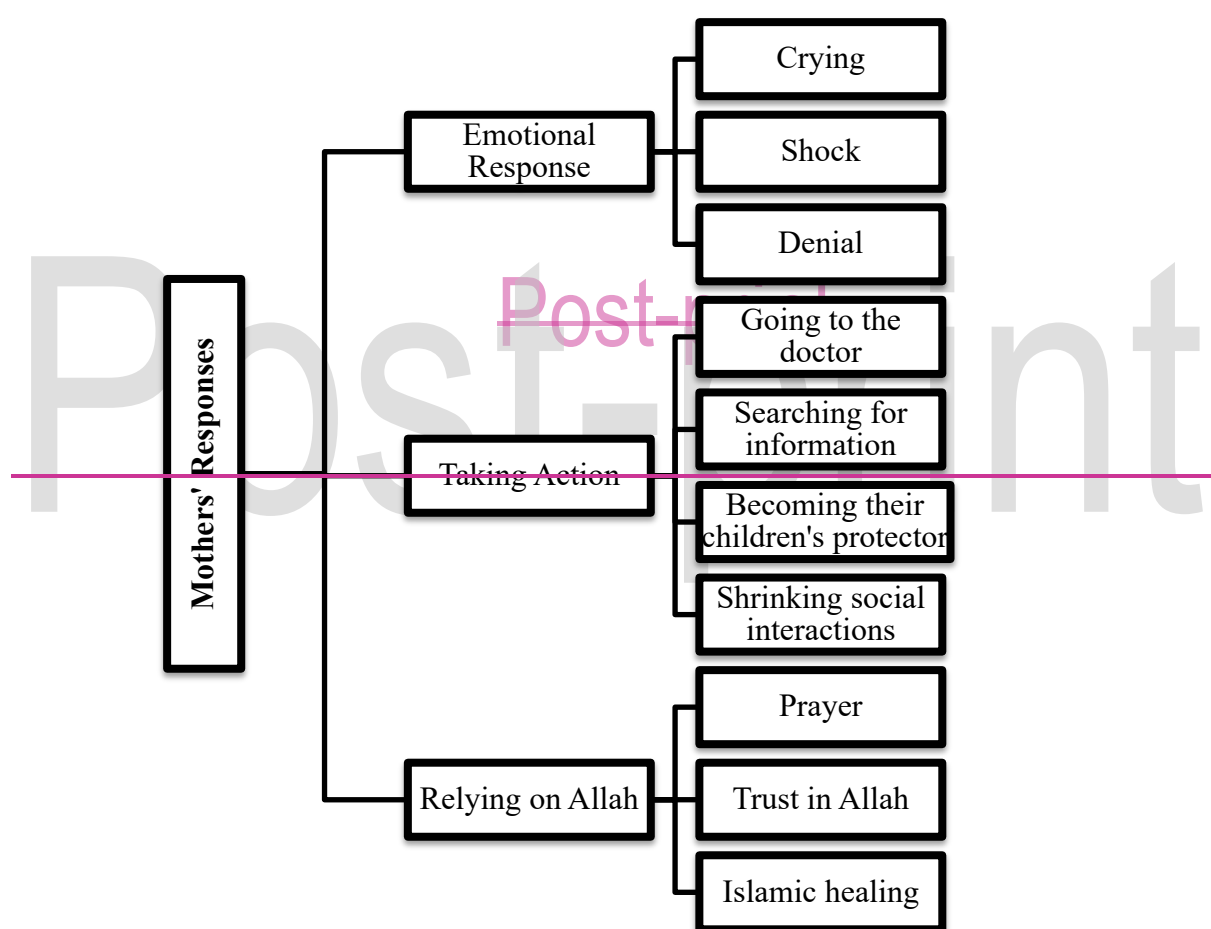


Image Description: Figure 1 provides a representation of the components of mothers' responses to shame. Their responses are divided into three major areas: Emotional Response, Taking Action, and Relying on Allah. Initial emotional responses mentioned by mothers included crying, shock, and denial. After their initial emotional response, some mothers described actions they took or experienced in response to their children's diagnosis. Such actions included going to the doctors, searching for information, becoming their children's protectors, and experiencing shrinking social interactions. Finally, for those mothers who reported relying on Allah, they responded with prayer, put their trust in Allah, or relied on

Islamic healing.

Most mothers interviewed noticed the challenges their children were facing at an early age. Nine of the 15 mothers interviewed, received a diagnosis of autism for their children between the ages of 1 and 2 years old. Of the 15 mothers, four learned of their children's condition when they were 3 years old. One of the mothers reported she learned of her child's condition when the child was 1 year old. Of the 15, only one mother learned of her child's condition at a later age, after the child was 6 years old.

Mothers' emotional responses

The mothers reacted differently to their children's diagnosis. Their initial responses reflected the emotional difficulty associated with receiving a diagnosis. Most of the mothers (M1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, and 14) agreed about the emotional challenges they faced shortly after their child's diagnosis. Their responses ranged from surprise to crying. Mother 3 reported that an overwhelming feeling of sadness overcame her. Mother 4 indicated she was in a state of disbelief. She said, "at first, of course, I was in shock, denial, and crying. I tried to deny it in every way possible" (M4-5). She could not do anything to stop the feeling and claimed that "the issue physically drained his father and me" (M4-6).

For some, however, they felt a sense of isolation. Mother 5 asserted that she was in shock and responded with "crying and isolation from everybody for several days. I became hot tempered and cried a lot" (M5-7). Some took this on themselves as Mother 5 explained that she "was in denial and stopped eating and drinking. My brain completely stopped thinking" (M5-8). However, she was not the only one who experienced this feeling. Mother 8 explained, "My husband and I got extremely sad and cried when we figured out that our child has autism. We were definitely shocked because he is our first child" (M8-6).

Isolation sometimes resulted in mothers trying to cover up the issue. Mother 9 said that she and her husband "got scared and did not tell anyone about her disorder" (M9-6). While this was a big problem for the parents, some had a harder time than others. Mother 12 made this point when she described the difficulty her husband had with accepting their child's diagnosis. She indicated, "I was so confused and couldn't believe that my son has autism. My husband and I had a tough time accepting it. I can't forget what my husband said: he can't [be] autistic, he can write, talk, and read. My son cannot be autistic! He is a normal child!" (M12-6). For some, the feelings took longer to deal with. Mother 14 explained that it took her "three months to get rid of that feeling. I felt extremely sad for his situation" (M14-7).

Taking action

Despite initial difficulty in dealing with the diagnosis, eventually some mothers were driven to take positive steps to help their children (M2, 3, 4, 6, 7, 9, 10, 11, 13, 14, and 15). Mother 2 reported, "I tried to take care of him more and visited specialists and experts including counselors and a private tutor specialized in these cases. I also followed up with hospitals and took him to rehabilitation sessions and massage therapy for exercising the affected muscles" (M2-6). This desire to seek help was echoed by others. Mother 7 posited that she "tried to search and read. I wrote the symptoms on the Internet to know what is wrong

with him because I had no knowledge about ASD and its symptoms. I asked a counselor. I visited an otolaryngologist. I attended educational workshops. I gained awareness and I learned about autism and autistic mood and thinking” (M7-6). Not only did mothers seek diagnosis, but some often went beyond that to learn more about their children’s conditions. Mother 10 emphasized this when she said, “I read a lot of articles from Google and I watched a lot of videos on YouTube about this disorder” (M10-7). Some invested money in acquiring more help. Mother 11 indicated that she “asked some speech specialists for some assistance to help my daughter talk” (M11-9).

These mothers also assumed the role of protector for their children. However, sometimes, they became overprotective. Mother 2 said, “I prefer to always be with him when we get out. I don’t like taking him with me to other people’s houses. When I do, I would be close to him the whole time because he does not defend himself” (M2-34).

Being overprotective was often designed to ensure the child’s feelings were not hurt by others. Mother 9 argued that she tried to be with her daughter to “avoid having others hurt her feelings” (M9-38). These protective mothers often worked to stop others from hurting their children. Mother 9 made this point when she described, “a year after my daughter’s diagnosis, I became stronger and I was able to confront them and stop them from making any negative comments because I did not want my daughter to get affected or hate her life when she grows up” (M9-41). Mother 15 agreed that she stopped anyone who tried to mock her child. This strength translated into trying to learn more about others. Mother 3 explained that after becoming overprotective, she felt “much better than before because I learned a lot about his disorder. I also tried to let my friends/relatives to understand/learn about my child’s condition” (M3-39). Mother 15 made the same point when she said, “I learned a lot about his condition, and I want the best for him. I tried to talk to his friends and told them about his disorder” (M15-21). These mothers were not only strong in advocating and standing up for their children, but they also worked hard to learn about the condition and educate others around them.

A shrinking social circle

Some mothers had to deal with a shrinking social circle that resulted from their children’s condition. Six of the 15 mothers (M1, 2, 5, 6, 8, and 9) reported issues with their shrinking social circles. Mother 1 explained that she was unhappy with some of the comments made by people in her circle, which led her to isolate herself. She posited, “I isolated myself and only relied on educated people who supported positively because at first, I swear to Allah, I got really tired of comments made by some people” (M1-42). Mother 5 responded in a similar manner to people in her circle who stigmatized her child. She explained, “I isolated myself and I stayed with my son and I ignored them, and I celebrated my success alone” (M5-47). This type of criticism led many to stay home and keep their children away from the rest of the family. Mother 9 made this point most clearly when she indicated, “sometimes I don’t like to take her with me to any parties or special occasions to avoid criticism and blame or mocking looks that might lead my daughter to have more complications” (M9-29). Mothers’ shrinking social circles were a result of the misconceptions of others and justified by a desire

to protect their children from others' judgmental looks and harsh remarks.

Relying on Allah: Role of religion

Despite the difficulty of their situations, many mothers found comfort in their faith. Mothers 1, 5, 6, and 12 emphasized the role their faith played in helping them become more comfortable with their children's diagnoses. Some reported their faith helped them accept their child's condition. Mother 1 explained, "The first and best resource is Allah's blessing and praying for him only. I was content with what Allah has chosen for me. It made me pleased with my child and I saw improvement. Thanks to Allah" (M1-37). For her, this reliance on Allah "helped and guided/showed me how to behave so I isolated myself indirectly and I felt comfortable" (M1-41). Furthermore, she went on to explain the role the Qur'an can play in helping heal autism. "I ask you and other brothers and sisters who dedicated themselves for this cause to pay attention to my viewpoint/opinion. I think autism can be healed through [the] Qur'an and it is related to satanic/demonic forces. And I have proof for that" (M1-56). For Mother 1, her reliance on Islamic healing helped her child tremendously. She claimed:

To be honest with you, I relied on Islamic healing, and I would recite three times a day and give them honey and ground black seeds and holy water. And the result was my older son improved [by] 80% (M1-58).

Similar to Mother 1, Mother 5 explained that her reliance on Allah made things easier after the initial challenges. She explained, "even his father abandoned him and refused to pay attention, but Allah with his mercy made things easy" (M5-9). Mother 6 found the same comfort in her submission to Allah, relying on prayer to help her child, viewing the condition as "a test from Allah" (M6-37). Because it was perceived as a test, the solution was in trusting God. Mother 12 emphasized that "the effect of trusting God is a miracle. I believe in leaving it to God and to trust him and to pray. Keeping that connection with God is the solution" (M12-41).

Consistent with the reliance on faith was the cultural belief that autism may be a result of an evil eye. Mother 11 explained that members of her family told her that this might be an evil eye. She explained, "two of my sisters-in-law told me that my daughter has an evil eye because she is so beautiful. They told me there are several ways to get rid of it: reading [the] Qur'an and taking her to a Sheikh" (M11-19).

Recommendations

Mothers in this study experienced social stigma similar to that proposed by Goffman (1963) and reiterated by Bos and colleagues (2013) and Olkin (2002). Others in their social circles blamed the mothers or their children for behaviors that deviated from what was perceived as society's norms. Mothers responded in a variety of ways that allowed them to cope with and in some cases, actively confront the stigma they and their children were experiencing.

This study's findings provide many implications for practice and demonstrate the importance of utilizing and making the most of the mothers' experiences. As the primary caregivers, mothers know more about their children than anyone else. While some specialists tend to ignore mothers' input and experiences, it is clear that a clear diagnosis and the determination of appropriate services should be responsive to mothers' wishes, to ensure services are culturally appropriate and sensitive to families' unique needs.

To overcome the problem of excluding mothers or having them feel as if they cannot bring their children into the community—for fear of stigmatization or negative reactions from others—specialists and other concerned individuals should develop individualized self-management systems to help children develop appropriate social behaviors. Self-management systems are not only useful for helping individuals with disabilities achieve certain behavioral goals but are also inclusive of the parents.

There are four stages when introducing a self-management plan. First, the collaborative team discusses diverse activities for effective and efficient implementation of a plan. The team should include teachers, specialists, and parents who identify and develop target behaviors they want to enhance or curb (Busick & Neitzel, 2009). One way to accomplish this is through goal setting. Goal setting involves coming up with a target for changing behavior (Schulze, 2016) and enables children with ASD to motivate themselves in addressing specific deficits in their abilities related to organizing, planning, and executive functioning.

Next, the self-management system must be effectively introduced to the learner. In this scenario, students should understand vital elements of the system, for instance, they should demonstrate the target behaviors and determine if key aspects of the behavior have taken place or not. The learner is engaged in recording the occurrence or lack of target behaviors. Students should effectively manage the reinforcements related to the achievement of identified goals (Busick & Neitzel, 2009). In this stage, students are encouraged to internalize the steps of the system in their own words to check for understanding. Using a self-instruction strategy is one way to measure comprehension which requires the child to verbalize the steps of a given task to be completed (Schulze, 2016). Self-statements prompt the child to promote a positive behavioral change and gives a student with ASD the ability to perform tasks independently.

The third stage in the process is implementation. Teachers provide the necessary materials for learners to utilize the self-management system as well as enable them to acquire necessary materials independently. Additionally, practitioners may provide different cues such as visual aids and verbal cues as well as teaching them how to self-record target behaviors. Teachers then educate learners on how to acquire access to earned reinforcements (Busick & Neitzel, 2009).

Finally, the fourth stage is to allow students to work independently and measure their development. Practitioners employ periodic check-ups to determine whether the learners are consistent in self-recording and reinforcement. The next step is for teachers to gradually

heighten specified criteria for success while also assessing the success of the students (Busick & Neitzel, 2009).

Another means of measuring outcomes is through self-charting, which empowers students to determine how effective they are in achieving their target goals. Self-charting works hand in hand with goal setting and self-monitoring to increase students' time on activities, academic performance, and accuracy. Furthermore, this strategy enables children with ASD to monitor their progress in behavioral change (Schulze, 2016).

The utility of this strategy lies in its collaborative nature (Busick & Neitzel, 2009). Teachers, parents, and education specialists work together to design plans that help learners achieve individualized behavioral goals, and this collaboration should take mothers' concerns into account. Although all of these steps are used in a school setting, mothers can and should participate in their implementation, learn how to introduce these strategies, and implement them on their own.

Peer support is also required to help both peers and individuals with ASD work in an environment that is inclusive. Studies have shown that peer supports consistently demonstrate increased levels of tolerance, awareness, and acceptance of differences (Bellini, Peters, Benner, & Hopf, 2007; Harper, Symon, & Frea, 2008; Kamps et al., 1992; Kohler, Greteman, Raschke, & Highnam, 2007). Similar support could be provided to family members. Researchers have shown that:

Family therapists are well-suited to help parents stay connected to each other as they create a "new normal." However, family therapists need updated information about autism, and they need to understand how family therapy can help parents of children with autism. Because having a child with autism affects multiple domains of family life...family therapists can utilize an integrative approach with parents, enabling them to flexibly work with the domains of action, meaning, and emotion (Solomon & Chung, 2012, p. 250).

It is also apparent that some mothers deal not only with stigma, but also other cultural beliefs that may inhibit their willingness to trust modern scientific approaches to disability. For specialists who work in this type of environment, the effort that is required is doubled, due to the fact that they must not only help mothers, but also be able to understand the underlying causes behind their fears and lack of trust. As some of these mothers demonstrate, belief in the supernatural as a cause of disability has led some of them to seek traditional methods for dealing with their children's conditions. Professionals should be aware of the need to adapt and to be flexible when working with individuals from different cultural backgrounds. As future specialists are trained, they must be aware of cultural differences. This also includes dealing with people from the same culture, as this study shows, it is important to remember that individuals' responses to disability vary depending on many factors. Al-Aoufi, Al-Zyoud, and Shaminan (2012) eloquently articulated this concern when they argued that:

Despite the fact that Muslims share the same beliefs and principles, people's attitudes and understanding regarding these concepts, and their reactions to individuals with a disability, may vary depending on the intensity of a person's faith, as well as their socio-economic status, level of education, awareness and, more importantly, their cultural context (p. 205).

Such an awareness is necessary in a world that has become incredibly interconnected and diverse in its makeup.

Clearly, much needs to be done to help increase awareness and acceptance of disability. On the institutional level, especially in schools, teachers need to receive proper training to increase their readiness to provide support for students with special needs, including those with ASD. Curricular activities ought to be redesigned to take these needs into account.

Finally, on a societal level, the adoption of a social model of disability is needed which "locates the disablement in the environment and society" (Olkin, 2002, p. 132), as opposed to the moral or medical models which place blame on the individual. One way to support this paradigm change is through the media which can play an instrumental role in helping increase awareness of different impairments, including autism spectrum disorder, and how society and the environment create disabling conditions. This could be achieved through educational programs that teach others about impairments and the subsequent disabling conditions. It also imperative to explore, "the mutual contribution television and disability studies can make to one another, focusing on the role of television and visual media in raising awareness and challenging gaps in understanding" of disabling conditions that are frequently stigmatized (Heward, Palfreman-Kay, & Innes, 2015, p. 229).

Conclusion

As noted above, many of these mothers had negative perceptions of their children's conditions and felt overwhelming sadness upon discovering that their children had ASD. Although these experiences and attitudes improved upon learning more about the disability, many still struggled with the cultural stigma that exists. And while many have found specialized institutions useful, much is needed to help improve people's awareness regarding impairments in an attempt to help the public be more understanding of individuals with ASD and more empathetic toward their experiences. Focusing on developing the knowledge and understanding of members of society in order to impact the present culture, only then might it be possible to address society's perceptions of impairments and eliminate the associated stigma experienced by individuals with impairments and those who love and support them.

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References

1. [Skoog I. Psychiatric disorders in the elderly. *Can J Psychiatry*. 2011;56\(7\):387-397.](#)
2. [Ferrari AJ, Charlson FJ, Norman RE, et al. Burden of depressive disorders by country, sex, age, and year: findings from the global burden of disease study 2010. *PLoS Med*. 2013;10\(11\):e1001547.](#)
3. [Djernes JK. Prevalence and predictors of depression in populations of elderly: a review. *Acta Psychiatr Scand*. 2006;113\(5\):372-387.](#)
4. [Luppa M, Sikorski C, Luck T, et al. Age- and gender-specific prevalence of depression in latest-life--systematic review and meta-analysis. *J Affect Disord*. 2012;136\(3\):212-221.](#)
5. [Barua A, Ghosh MK, Kar N, Basilio MA. Prevalence of depressive disorders in the elderly. *Ann Saudi Med*. 2011;31\(6\):620-624.](#)
6. [Karlsson B, Johnell K, Sigstrom R, Sjöberg L, Fratiglioni L. Depression and Depression Treatment in a Population-Based Study of Individuals Over 60 Years Old Without Dementia. *Am J Geriatr Psychiatry*. 2016;24\(8\):615-623.](#)
7. [Castro-Costa E, Dewey M, Stewart R, et al. Prevalence of depressive symptoms and syndromes in later life in ten European countries: the SHARE study. *Br J Psychiatry*. 2007;191:393-401.](#)
8. [Fiske A, Wetherell JL, Gatz M. Depression in older adults. *Annu Rev Clin Psychol*. 2009;5:363-389.](#)
9. [Vaupel JW. Biodemography of human ageing. *Nature*. 2010;464\(7288\):536-542.](#)
10. [Kontis V, Bennett JE, Mathers CD, Li G, Foreman K, Ezzati M. Future life expectancy in 35 industrialised countries: projections with a Bayesian model ensemble. *Lancet*. 2017;389\(10076\):1323-1335.](#)
11. [Vos T, Allen C, Arora M, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990-2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet*. 2016;388\(10053\):1545-1602.](#)
12. [Kendrick T, Stuart B, Newell C, Geraghty AW, Moore M. Changes in rates of recorded depression in English primary care 2003-2013: Time trend analyses of effects of the economic recession, and the GP contract quality outcomes framework \(QOF\). *J Affect Disord*. 2015;180:68-78.](#)
13. [Mojtabai R, Olfson M, Han B. National Trends in the Prevalence and Treatment of Depression in Adolescents and Young Adults. *Pediatrics*. 2016;138\(6\).](#)
14. [Jeuring HW, Comijs HC, Deeg DJH, Stek ML, Huisman M, Beekman ATF. Secular trends in the prevalence of major and subthreshold depression among 55-64-year olds over 20 years. *Psychol Med*. 2017;1-11.](#)
15. [Hawthorne G, Goldney R, Taylor AW. Depression prevalence: is it really increasing? *Aust N Z J Psychiatry*. 2008;42\(7\):606-616.](#)
16. [Wiberg P, Waern M, Billstedt E, Ostling S, Skoog I. Secular trends in the prevalence of dementia and depression in Swedish septuagenarians 1976-2006. *Psychol Med*. 2013;43\(12\):2627-2634.](#)

17. [Keyes KM, Nicholson R, Kinley J, et al. Age, period, and cohort effects in psychological distress in the United States and Canada. *Am J Epidemiol*. 2014;179\(10\):1216-1227.](#)
18. [Spiers N, Brugha TS, Bebbington P, McManus S, Jenkins R, Meltzer H. Age and birth cohort differences in depression in repeated cross-sectional surveys in England: the National Psychiatric Morbidity Surveys, 1993 to 2007. *Psychol Med*. 2012;42\(10\):2047-2055.](#)
19. [Vink D, Aartsen MJ, Schoevers RA. Risk factors for anxiety and depression in the elderly: A review. *J Affect Disord*. 2008;106\(1-2\):29-44.](#)
20. [Mac Giolla E, Kajonius PJ. Sex differences in personality are larger in gender equal countries: Replicating and extending a surprising finding. *Int J Psychol*. 2018.](#)
21. [Duberstein PR, Palsson SP, Waern M, Skoog I. Personality and risk for depression in a birth cohort of 70-year-olds followed for 15 years. *Psychol Med*. 2008;38\(5\):663-671.](#)
22. [Twenge JM. The age of anxiety? Birth cohort change in anxiety and neuroticism, 1952-1993. *J Pers Soc Psychol*. 2000;79\(6\):1007-1021.](#)
23. [Billstedt E, Waern M, Duberstein P, et al. Secular changes in personality: study on 75-year-olds examined in 1976-1977 and 2005-2006. *Int J Geriatr Psychiatry*. 2013;28\(3\):298-304.](#)
24. [Roberts BW, Walton KE, Viechtbauer W. Patterns of mean-level change in personality traits across the life course: a meta-analysis of longitudinal studies. *Psychol Bull*. 2006;132\(1\):1-25.](#)
25. [Billstedt E, Skoog I, Duberstein P, et al. A 37-year prospective study of neuroticism and extraversion in women followed from mid-life to late life. *Acta Psychiatr Scand*. 2014;129\(1\):35-43.](#)
26. [Sacuiu S, Gustafson D, Sjogren M, et al. Secular changes in cognitive predictors of dementia and mortality in 70-year-olds. *Neurology*. 2010;75\(9\):779-785.](#)
27. [Lak VW, Skoog I, Guo X. Secular trends in lung function and its relation to survival in Swedish 75 year olds 1976-2006. *Age Ageing*. 2012;41\(6\):735-740.](#)
28. [Zhi X, Joas E, Waern M, Ostling S, Borjesson-Hanson A, Skoog I. Prevalence of cardiovascular disorders and risk factors in two 75-year-old birth cohorts examined in 1976-1977 and 2005-2006. *Aging Clin Exp Res*. 2013;25\(4\):377-383.](#)
29. [Beckman N, Waern M, Ostling S, Sundh V, Skoog I. Determinants of sexual activity in four birth cohorts of Swedish 70-year-olds examined 1971-2001. *J Sex Med*. 2014;11\(2\):401-410.](#)
30. [Skoog I, Borjesson-Hanson A, Kern S, et al. Decreasing prevalence of dementia in 85-year olds examined 22 years apart: the influence of education and stroke. *Sci Rep*. 2017;7\(1\):6136.](#)
31. [Wu YT, Beiser AS, Breteler MMB, et al. The changing prevalence and incidence of dementia over time - current evidence. *Nat Rev Neurol*. 2017;13\(6\):327-339.](#)
32. [Dorius SF, Firebaugh G. Trends in Global Gender Inequality. *Soc Forces*. 2010;88\(5\):1941-1968.](#)
33. [Rinder L, Roupe S, Steen B, Svanborg A. Seventy-year-old people in Gothenburg. A population study in an industrialized Swedish city. *Acta Med Scand*. 1975;198\(5\):397-407.](#)
34. [Nilsson LV. Prevalence of mental disorders in a 70-year-old urban sample - a cohort comparison. *J Clinical Exp Gerontol*. 1983;5\(2\):101-120.](#)
35. [Bengtsson C, Ahlqvist M, Andersson K, Bjorkelund C, Lissner L, Soderstrom M. The Prospective Population Study of Women in Gothenburg, Sweden, 1968-69 to 1992-93. A 24-year follow-up study with special reference to participation, representativeness, and mortality. *Scand J Prim Health Care*. 1997;15\(4\):214-219.](#)
36. [Rydberg Sterner T, Ahlner F, Blennow K, et al. The Gothenburg H70 Birth cohort study 2014-16: design, methods and study population. *Eur J Epidemiol*. 2018; doi: 10.1007/s10654-018-0459-8. \[Epub ahead of print\].](#)
37. [McHugh ML. Interrater reliability: the kappa statistic. *Biochem Med \(Zagreb\)*. 2012;22\(3\):276-282.](#)
38. [Asberg M, Schalling D. Construction of a New Psychiatric Rating Instrument, the Comprehensive Psychopathological Rating-Scale \(CPRS\). *Prog Neuropsychopharmacol*. 1979;3\(4\):405-412.](#)

39. [van der Laan NC, Schimmel A, Heeren TJ. The applicability and the inter-rater reliability of the Comprehensive Psychopathological Rating Scale in an elderly clinical population. *Int J Geriatr Psychiatry*. 2005;20\(1\):35-40.](#)
40. [Montgomery SA, Asberg M. A new depression scale designed to be sensitive to change. *Br J Psychiatry*. 1979;134:382-389.](#)
41. [Mottram P, Wilson K, Copeland J. Validation of the Hamilton Depression rating scale and Montgomery and Asberg Rating Scales in terms of AGE-CAT depression cases. *Int J Geriatr Psychiatry*. 2000;15\(12\):1113-1119.](#)
42. [Eysenck HJ & Eysenck SBJ. Manual of the Eysenck Personality Questionnaire. Hodder & Stoughton; 1975.](#)
43. [Skoog I, Nilsson L, Landahl S, Steen B. Mental disorders and the use of psychotropic drugs in an 85-year-old urban population. *Int Psychogeriatr*. 1993;5\(1\):33-48.](#)
44. [American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th ed. Washington, DC: American Psychiatric Association; 2013.](#)
45. [American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders: DSM-IV-TR. 4th ed. Washington, DC: American Psychiatric Association; 2000.](#)
46. [Kay DW, Roth M, Beamish P. Old Age Mental Disorders in Newcastle Upon Tyne. II. A Study of Possible Social and Medical Causes. *Br J Psychiatry*. 1964;110:668-682.](#)
47. [American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 3rd ed. \(revised\). Washington, DC: American Psychiatric Association; 1987.](#)
48. [Wancata J, Borjesson-Hanson A, Ostling S, Sjogren K, Skoog I. Diagnostic criteria influence dementia prevalence. *Am J Geriatr Psychiatry*. 2007;15\(12\):1034-1045.](#)
49. [Nordic Council on Medicines. Nordic statistics on medicines 1981-1983. Guidelines for ATC classification. 14th vol. NLN publications; 1986.](#)
50. [Schmitt DP, Realo A, Voracek M, Allik J. Why can't a man be more like a woman? Sex differences in big five personality traits across 55 cultures. *J Pers Soc Psychol*. 2008;94\(1\):168-182.](#)
51. [Skoog I. Dementia incidence - the times, they are a-changing. *Nature Reviews Neurology*. 2016;12\(6\):316-U315.](#)
52. [Dorius SF, Firebaugh G. Trends in Global Gender Inequality \(Forthcoming, Social Forces\). *Soc Forces*. 2010;88\(5\):1941-1968.](#)
53. [Piccinelli M, Wilkinson G. Gender differences in depression. Critical review. *Br J Psychiatry*. 2000;177:486-492.](#)
54. [Kasen S, Cohen P, Berenson K, Chen H, Dufur R. Dual work and family roles and depressive symptoms in two birth cohorts of women. *Soc Psychiatry Psychiatr Epidemiol*. 2005;40\(4\):300-307.](#)
55. [Nolen-Hoeksema S. Emotion Regulation and Psychopathology: The Role of Gender. *Annu Rev Clin Psycho*. 2012;8:161-187.](#)
56. [Angst J, Dobler-Mikola A. Do the diagnostic criteria determine the sex ratio in depression? *J Affect Disord*. 1984;7\(3-4\):189-198.](#)
57. [Naninck EF, Lucassen PJ, Bakker J. Sex differences in adolescent depression: do sex hormones determine vulnerability? *J Neuroendocrinol*. 2011;23\(5\):383-392.](#)
58. [Hodes GE, Walker DM, Labonte B, Nestler EJ, Russo SJ. Understanding the epigenetic basis of sex differences in depression. *J Neurosci Res*. 2017;95\(1-2\):692-702.](#)
59. [van Loo HM, Aggen SH, Gardner CO, Kendler KS. Sex similarities and differences in risk factors for recurrence of major depression. *Psychol Med*. 2018;48\(10\):1685-1693.](#)
60. [Carstensen LL, Turan B, Scheibe S, et al. Emotional Experience Improves With Age: Evidence Based on Over 10 Years of Experience Sampling. *Psychol Aging*. 2011;26\(1\):21-33.](#)
61. [Sjoberg L, Karlsson B, Atti AR, Skoog I, Fratiglioni L, Wang HX. Prevalence of depression: Comparisons of different depression definitions in population-based samples of older adults. *J Affect Disord*. 2017;221:123-131.](#)

62. [Markkula N, Zitko P, Pena S, Margozzini P, Retamal CP. Prevalence, trends, correlates and treatment of depression in Chile in 2003 to 2010. Soc Psychiatry Psychiatr Epidemiol. 2017;52\(4\):399-409.](#)
63. [Collishaw S. Annual research review: Secular trends in child and adolescent mental health. J Child Psychol Psychiatry. 2015;56\(3\):370-393.](#)
64. [Kessler RC, McGonagle KA, Nelson CB, Hughes M, Swartz M, Blazer DG. Sex and depression in the National Comorbidity Survey. II: Cohort effects. J Affect Disord. 1994;30\(1\):15-26.](#)
65. [Patten SB. The prevalence of major depression is not changing. Can J Psychiatry. 2015;60\(1\).](#)
66. [Compton WM, Conway KP, Stinson FS, Grant BF. Changes in the prevalence of major depression and comorbid substance use disorders in the United States between 1991-1992 and 2001-2002. Am J Psychiatry. 2006;163\(12\):2141-2147.](#)
67. [Kessler RC, Demler O, Frank RG, et al. Prevalence and treatment of mental disorders, 1990 to 2003. N Engl J Med. 2005;352\(24\):2515-2523.](#)
68. [Murphy JM, Laird NM, Monson RR, Sobol AM, Leighton AH. A 40-year perspective on the prevalence of depression: the Stirling County Study. Arch Gen Psychiatry. 2000;57\(3\):209-215.](#)
69. [Wickramaratne PJ, Weissman MM, Leaf PJ, Holford TR. Age, period and cohort effects on the risk of major depression: results from five United States communities. J Clin Epidemiol. 1989;42\(4\):333-343.](#)
70. [Klerman GL, Lavori PW, Rice J, et al. Birth-cohort trends in rates of major depressive disorder among relatives of patients with affective disorder. Arch Gen Psychiatry. 1985;42\(7\):689-693.](#)
71. [Weissman MM, Wickramaratne P, Greenwald S, et al. The Changing Rate of Major Depression - Cross-National Comparisons. Jama-J Am Med Assoc. 1992;268\(21\):3098-3105.](#)
72. [Murphy JM, Laird NM, Monson RR, Sobol AM, Leighton AH. Incidence of depression in the Stirling County Study: historical and comparative perspectives. Psychol Med. 2000;30\(3\):505-514.](#)
73. [Mattisson C, Bogren M, Nettelbladt P, Munk-Jorgensen P, Bhugra D. First incidence depression in the Lundby Study: a comparison of the two time periods 1947-1972 and 1972-1997. J Affect Disord. 2005;87\(2-3\):151-160.](#)
74. [Statistics Sweden. Life expectancy 1751–2017.
https://www.scb.se/en/finding-statistics/statistics-by-subject-area/population/population-composition/population-statistics/pong/tables-and-graphs/yearly-statistics--the-whole-country/life-expectancy/. Accessed 2018-06-26.](#)
75. [Klein DN, Kotov R, Bufferd SJ. Personality and depression: explanatory models and review of the evidence. Annu Rev Clin Psychol. 2011;7:269-295.](#)
76. [de Moor MH, van den Berg SM, et al. Meta-analysis of Genome-wide Association Studies for Neuroticism, and the Polygenic Association With Major Depressive Disorder. Jama Psychiat. 2015;72\(7\):642-650.](#)
77. [Kotov R, Gamez W, Schmidt F, Watson D. Linking "Big" Personality Traits to Anxiety, Depressive, and Substance Use Disorders: A Meta-Analysis. Psychol Bull. 2010;136\(5\):768-821.](#)
78. [Patten SB, Williams JV, Lavorato DH, Fiest KM, Bulloch AG, Wang J. Antidepressant use in Canada has stopped increasing. Can J Psychiatry. 2014;59\(11\):609-614.](#)
79. [Sonnenberg CM, Deeg DJ, Comijs HC, van Tilburg W, Beekman AT. Trends in antidepressant use in the older population: results from the LASA-study over a period of 10 years. J Affect Disord. 2008;111\(2-3\):299-305.](#)
80. [Isacsson G, Ahlner J. Antidepressants and the risk of suicide in young persons--prescription trends and toxicological analyses. Acta Psychiatr Scand. 2014;129\(4\):296-302.](#)
81. [Isacsson G, Rich CL, Jureidini J, Raven M. The increased use of antidepressants has contributed to the worldwide reduction in suicide rates. Br J Psychiatry. 2010;196\(6\):429-433.](#)

1301 82. [Giuffra LA, Risch N. Diminished recall and the cohort effect of major depression: a simulation](#)
1302 [study. Psychol Med. 1994;24\(2\):375-383.](#)

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TABLE 1. Sample characteristics of 70-year-olds by sex and examination year

Examination year Birth cohort	Men			Women			
	1976-77 1906-07	2000-02 1930	2014-16 1944	1976-77 1906-07	1992-93 1922	2000-02 1930	2014-16 1944
	% (no. of cases/ total cases)	% (no. of cases/ total cases)	% (no. of cases/ total cases)	% (no. of cases/ total cases)	% (no. of cases/ total cases)	% (no. of cases/ total cases)	% (no. of cases/ total cases)
Dementia	<u>1.7 (3/177)</u>	<u>0.9 (2/229)</u>	<u>1.8 (10/552)</u>	<u>1.8 (4/222)</u>	<u>3.4 (8/234)</u>	<u>2.3 (6/266)</u>	<u>1.6 (10/634)</u>
N (dementia free sample)	<u>174</u>	<u>227</u>	<u>542</u>	<u>218</u>	<u>226</u>	<u>260</u>	<u>624</u>
Demographics							
> Compulsory education	<u>17.9 (31/173)^{§¶}</u>	<u>43.4 (98/226)[¶]</u>	<u>81.1 (438/540)</u>	<u>17.6 (38/214)^{‡§¶}</u>	<u>38.5 (79/205)[¶]</u>	<u>38.3 (98/256)[¶]</u>	<u>85.7 (533/622)</u>
Living alone	<u>0.6 (1/174)[¶]</u>	<u>‡</u>	<u>27.2 (145/533)</u>	<u>1.8 (4/218)[¶]</u>	<u>‡</u>	<u>‡</u>	<u>43.2 (266/616)</u>
Sheltered accommodation	<u>0.6 (1/172)</u>	<u>0 (0/227)</u>	<u>0.2 (1/538)</u>	<u>0 (0/218)</u>	<u>‡</u>	<u>0.4 (1/256)</u>	<u>0.3 (2/621)</u>
Relationships							
Lost partner during last 5 years	<u>4.6 (8/174)</u>	<u>1.3 (3/227)</u>	<u>4.1 (22/542)</u>	<u>11.0 (24/218)[¶]</u>	<u>11.9 (27/226)[¶]</u>	<u>7.3 (19/260)</u>	<u>5.1 (32/624)</u>
Having partner	<u>78.0 (135/173)</u>	<u>80.6 (183/227)</u>	<u>76.4 (407/533)</u>	<u>43.6 (95/218)^{‡¶}</u>	<u>54.2 (122/225)</u>	<u>50.8 (132/260)[¶]</u>	<u>58.8 (362/616)</u>
Happy relationship	<u>20.1 (35/174)^{§¶}</u>	<u>48.9 (111/227)</u>	<u>45.3 (244/539)</u>	<u>13.8 (30/218)^{‡§¶}</u>	<u>31.2 (68/218)</u>	<u>27.6 (71/257)</u>	<u>33.4 (205/614)</u>
Having children	<u>75.1 (130/173)^{§¶}</u>	<u>88.6 (186/210)</u>	<u>86.9 (469/540)</u>	<u>67.0 (146/218)^{‡§¶}</u>	<u>85.5 (183/214)</u>	<u>86.1 (211/245)</u>	<u>86.3 (537/622)</u>
Feeling lonely	<u>15.8 (27/171)</u>	<u>17.5 (37/211)[¶]</u>	<u>10.4 (55/530)</u>	<u>21.6 (47/218)[§]</u>	<u>‡</u>	<u>32.2 (79/245)[¶]</u>	<u>22.0 (134/609)</u>
Health							
Antidepressant use ^{††}	<u>2.3 (4/174)</u>	<u>3.1 (7/227)</u>	<u>5.9 (32/542)</u>	<u>5.5 (12/218)^{§¶}</u>	<u>8.0 (18/226)[¶]</u>	<u>13.5 (35/260)</u>	<u>13.1 (82/624)</u>
Back pain	<u>21.3 (37/174)[§]</u>	<u>36.7 (81/221)[¶]</u>	<u>28.5 (154/541)</u>	<u>37.6 (82/218)[§]</u>	<u>44.0 (99/225)</u>	<u>50.4 (127/252)[¶]</u>	<u>40.0 (248/620)</u>
Joint pain	<u>12.6 (22/174)^{§¶}</u>	<u>28.4 (63/222)[¶]</u>	<u>36.1 (194/537)</u>	<u>29.4 (64/218)^{§¶}</u>	<u>35.6 (80/225)^{§¶}</u>	<u>48.2 (122/253)</u>	<u>52.2 (320/613)</u>
Visit healthcare during last 3 months	<u>40.8 (71/174)</u>	<u>46.5 (94/202)</u>	<u>‡</u>	<u>52.8 (114/216)</u>	<u>58.8 (124/211)</u>	<u>60.0 (141/235)</u>	<u>‡</u>
Visit healthcare during last year	<u>64.9 (113/174)[¶]</u>	<u>‡</u>	<u>91.5 (495/541)</u>	<u>79.6 (172/216)[¶]</u>	<u>‡</u>	<u>‡</u>	<u>92.1 (568/617)</u>

All sample characteristics (dementia, demographics, relationships and health) have been tested for differences in proportions between examination years. Analyses of demographics, relationships, and health were based

on the dementia free sample.

† Data not available for this birth cohort.

†† Antidepressants (N06A) were classified according to the Anatomical Therapeutic Chemical (ATC) classification system recommended by the WHO.

‡ Significant difference compared to examination year 1992-93 ($p < 0.05$)

§ Significant difference compared to examination year 2000-02 ($p < 0.05$)

¶ Significant difference compared to examination year 2014-16 ($p < 0.05$)

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TABLE 2. Mean score of depressive symptoms and neuroticism by depression, sex and examination year/birth cohort

	Total sample				Participants with any depression				Participants without depression			
Examination year	1976-77	1992-93	2000-02	2014-16	1976-77	1992-93	2000-02	2014-16	1976-77	1992-93	2000-02	2014-16
Birth cohort	1906-07	1922	1930	1944	1906-07	1922	1930	1944	1906-07	1922	1930	1944
MADRS score[†], mean^(sd)												
N (men/women)	(174/218)	(†/226)	(227/260)	(542/624)	(11/33)	(†/41)	(21/44)	(36/68)	(163/185)	(†/185)	(206/216)	(506/556)
All	4.8 (5.8)¶	†	4.6 (5.5)¶	4.0 (5.1)	16.5 (8.3)	†	15.1 (6.8)	15.4 (7.2)	3.3 (3.1)	†	3.1 (3.0)	2.9 (3.0)
Men	3.6 (4.4)	†	3.9 (4.7)	3.6 (4.6)	14.6 (7.0)	†	14.3 (6.0)	14.6 (7.7)	2.8 (3.0)	†	2.8 (3.0)	2.8 (3.0)
Women	5.8 (6.5)¶	6.1 (7.3)¶	5.4 (6.1)	4.4 (5.4)	17.1 (8.7)	17.8 (8.4)	15.5 (7.1)	15.8 (6.9)	3.8 (3.0)¶	3.5 (3.5)	3.4 (3.0)	3.0 (3.1)
Neuroticism, mean^(sd)												
N (men/women)	(151/178)	(†/194)	†	(520/602)	(10/26)	(†/31)	†	(33/63)	(141/152)	(†/163)	†	(487/539)
All	7.8 (4.9)¶	†	†	6.1 (3.9)	12.5 (4.9)	†	†	10.8 (4.4)	7.2 (4.6)¶	†	†	5.6 (3.6)
Men	6.1 (4.1)	†	†	5.5 (3.8)	8.9 (4.6)	†	†	10.8 (4.8)	5.9 (4.0)¶	†	†	5.1 (3.5)
Women	9.2 (5.0)§¶	7.7 (4.1)¶	†	6.6 (3.9)	13.9 (4.4)¶	12.1 (3.6)	†	10.8 (4.3)	8.4 (4.7)§¶	6.8 (3.7)	†	6.1 (3.6)

† Data not available for this birth cohort.

‡ Depressive symptoms measured with Montgomery Åsberg Depression Rating Scale (MADRS)

§ Significant difference compared to examination year 1992-93 ($p < 0.05$)

¶ Significant difference compared to examination year 2014-16 ($p < 0.05$)

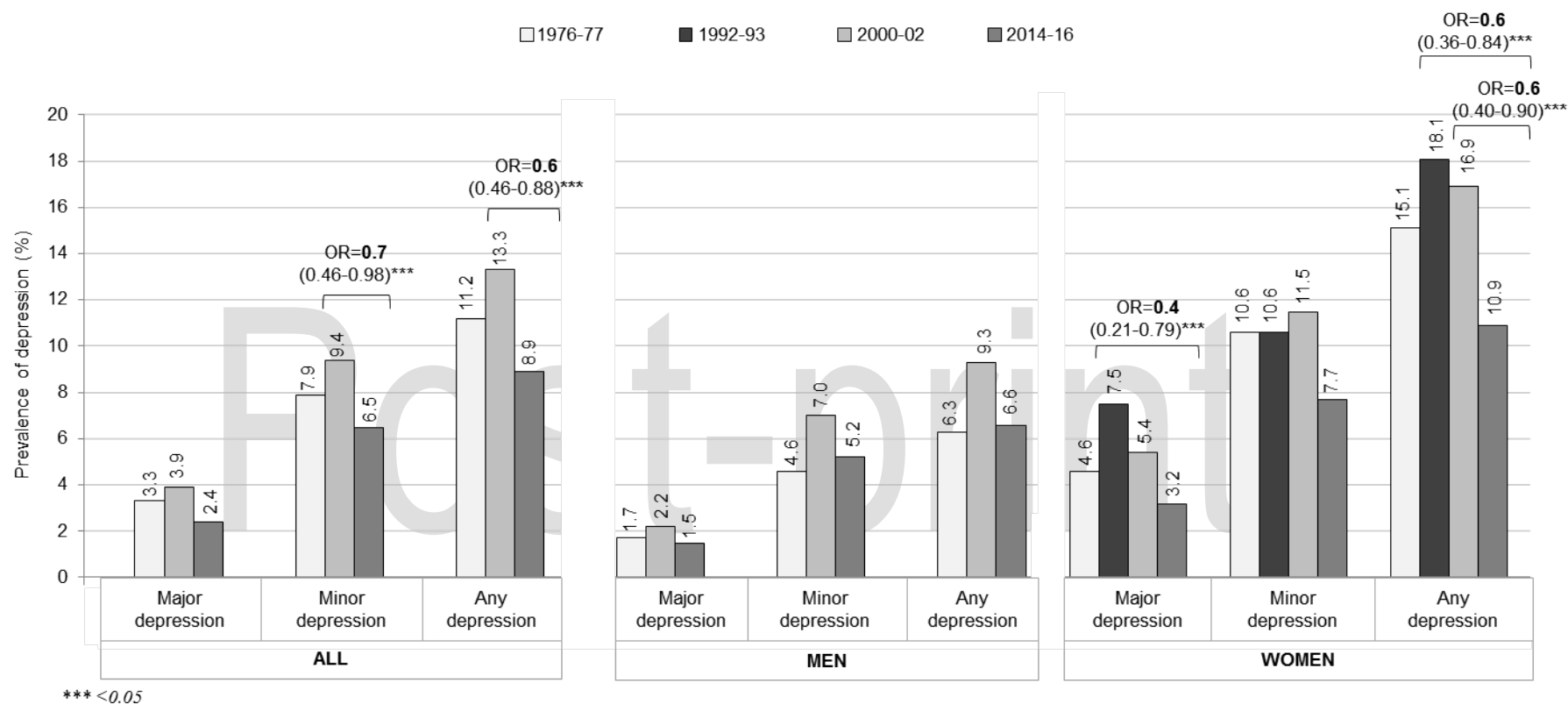


Figure 1. One-month prevalence of major, minor and any depression by sex and examination year:

*** < 0.05

OR = Odds Ratio

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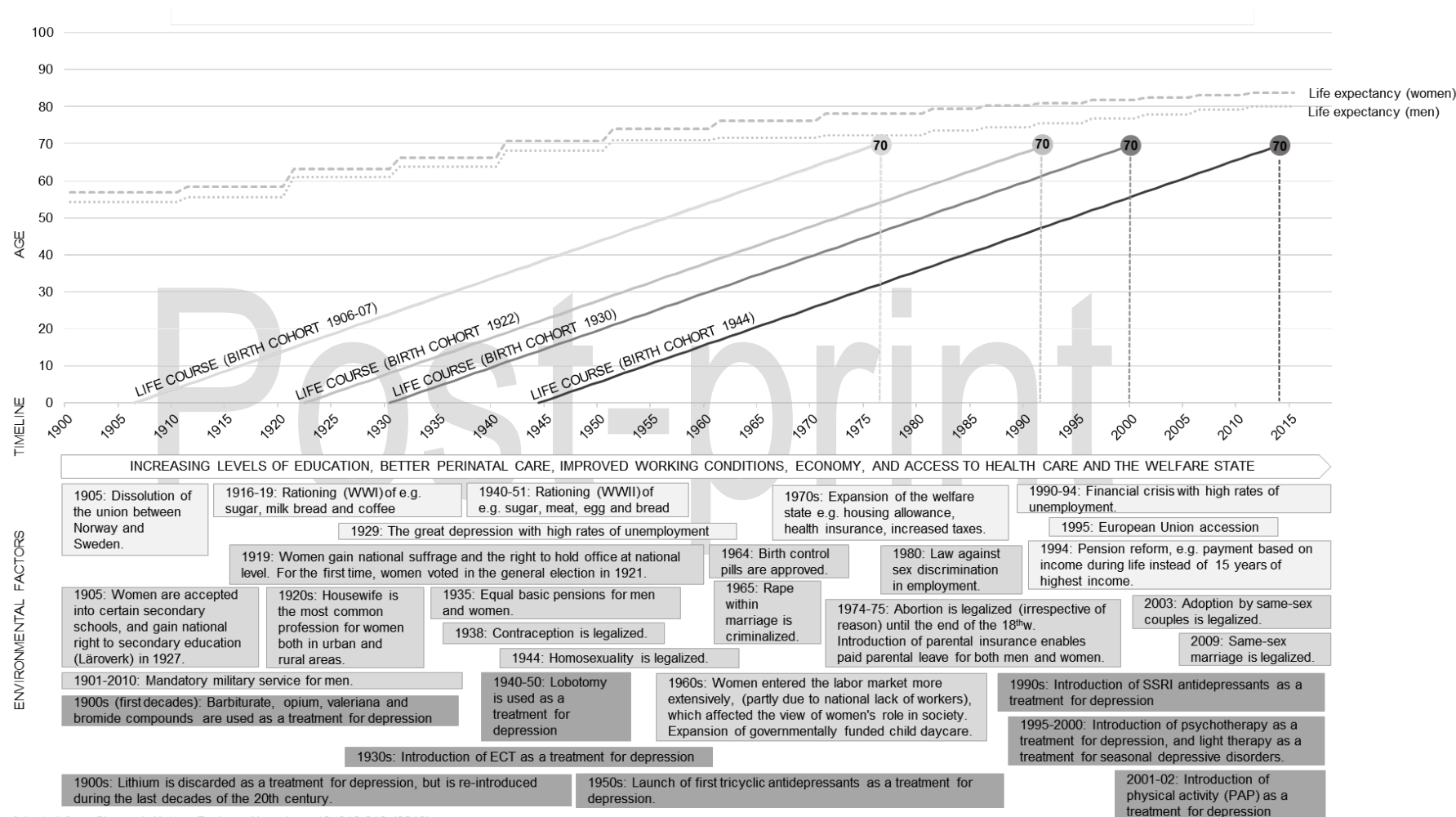
TABLE 3. Sex differences in depression and neuroticism in 70-year-olds by examination year/birth cohort

Examination year Birth cohort	1976-77 1906-07		2000-02 1930		2014-16 1944	
	<u>Women vs Men</u>	<u>p</u>	<u>Women vs Men</u>	<u>p</u>	<u>Women vs Men</u>	<u>p</u>
Total sample						
MADRS score (mean)	5.8/3.6	***	5.4/3.9	***	4.4/3.6	***
Neuroticism score (mean)	9.2/6.1	***	†	†	6.6/5.5	***
Participants without depression						
MADRS score (mean)	3.8/2.8	***	3.4/2.8	0.05	3.0/2.8	0.20
Neuroticism score (mean)	8.4/5.9	***	†	†	6.1/5.1	***
Participants with major depression						
No. of cases with major depression	(10/3)		(14/5)		(20/8)	
OR, (95 % CI)	2.7 (0.7-10.1)	0.13	2.5 (0.9-7.1)	0.08	2.2 (1.0-5.1)	0.06
MADRS score (mean)	26.9/19.3	0.12	21.6/20.2	0.65	23.8/26.6	0.17
Neuroticism score (mean)	17.1/8.0	***	†	†	11.4/13.3	0.30
Participants with minor depression						
No. of cases with minor depression	(23/8)		(30/16)		(48/28)	
OR, (95 % CI)	2.5 (1.1-5.7)	***	1.7 (0.9-3.2)	0.09	1.5 (0.9-2.5)	0.08
MADRS score (mean)	12.8/12.9	0.97	12.6/12.4	0.91	12.4/11.2	0.23
Neuroticism score (mean)	11.8/9.3	0.19	†	†	10.6/10.1	0.70
Participants with any depression						
No. of cases with any depression	(33/11)		(44/21)		(68/36)	
OR, (95 % CI)	2.6 (1.3-5.4)	***	2.0 (1.1-3.5)	***	1.7 (1.1-2.6)	***
MADRS score (mean)	17.1/14.6	0.41	15.5/14.3	0.50	15.8/14.6	0.44
Neuroticism score (mean)	13.9/8.9	***	†	†	10.8/10.8	0.98

† Data not available for this birth cohort.

*** $p < 0.05$

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Adapted from Skoog, I. Nature Reviews Neurology 12, 316-318 (2016)
Additional sources: Statistics Sweden 2018. Women and men in Sweden 2018 – Facts and Figures.

Figure 2. Historical context over the life course for cohorts born 1906-07, 1922, 1930, and 1944 in Sweden:

Adapted from Skoog I. Nature Reviews Neurology 12, 316-318 (2016) ⁵¹.

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Figure Legends

Figure 1. One-month prevalence of major, minor and any depression by sex and examination year:

*** < 0.05

OR = Odds Ratio

Figure 2. Historical context over the life course for cohorts born 1906-07, 1922, 1930, and 1944 in Sweden:

Adapted from Skoog I. Nature Reviews Neurology 12, 316-318 (2016) ⁵¹.

Supplementary Table 1. The association[†] between mean neuroticism score[‡] and depression by sex and examination year

Examination year Birth cohort	1976-77 1906-07					1992-93 1922					2014-16 1944				
	OR	SE	df	p	(95 % CI)	OR	SE	df	p	(95 % CI)	OR	SE	df	p	(95 % CI)
Any depression															
All	1.24	0.04	1	<0.01	(1.15-1.34)	§	§	§	§	§	1.33	0.03	1	<0.01	(1.26-1.41)
Men	1.17	0.08	1	0.035	(1.01-1.36)	§	§	§	§	§	1.34	0.04	1	<0.01	(1.23-1.46)
Women	1.26	0.05	1	<0.01	(1.14-1.39)	1.42	0.06	1	<0.01	(1.25-1.61)	1.32	0.04	1	<0.01	(1.23-1.42)
Major depression															
All	1.36	0.07	1	<0.01	(1.19-1.55)	§	§	§	§	§	1.34	0.05	1	<0.01	(1.23-1.47)
Men	1.11	0.13	1	0.43	(0.86-1.43)	§	§	§	§	§	1.38	0.08	1	<0.01	(1.19-1.59)
Women	1.50	0.10	1	<0.01	(1.22-1.83)	1.39	0.08	1	<0.01	(1.19-1.63)	1.32	0.06	1	<0.01	(1.18-1.48)
Minor depression															
All	1.15	0.04	1	0.01	(1.05-1.25)	§	§	§	§	§	1.27	0.03	1	<0.01	(1.21-1.35)
Men	1.19	0.09	1	0.05	(1.00-1.41)	§	§	§	§	§	1.27	0.04	1	<0.01	(1.17-1.39)
Women	1.12	0.05	1	0.03	(1.01-1.24)	1.27	0.06	1	<0.01	(1.12-1.44)	1.27	0.04	1	<0.01	(1.18-1.36)
	R ²	B (SE)	df	p	(95 % CI)	R ²	B (SE)	df	p	(95 % CI)	R ²	B (SE)	df	p	(95 % CI)
MADRS score															
All	0.23	0.6 (0.06)	1	<0.01	(0.47-0.70)	§	§	§	§	§	0.27	0.7 (0.03)	1	<0.01	(0.59-0.72)
Men	0.09	0.3 (0.09)	1	<0.01	(0.16-0.51)	§	§	§	§	§	0.27	0.6 (0.04)	1	<0.01	(0.53-0.71)
Women	0.27	0.7 (0.09)	1	<0.01	(0.52-0.87)	0.30	0.9 (0.10)	1	<0.01	(0.73-1.14)	0.26	0.7 (0.05)	1	<0.01	(0.59-0.77)

† GLM (independent variable=neuroticism)

‡ Mean neuroticism score was not available for the examination in 2000-02 (birth cohort 1930).

§ Data not available (only women were examined in 1992-93 (birth cohort 1922))

- Al-Aoufi, H., Al-Zyoud, N., & Shahminan, N. (2012). Islam and the cultural conceptualization of disability. *International Journal of Adolescence and Youth*, 17(4), 205–219.
- Alqahtani, M. (2012). Understanding autism in Saudi Arabia: A qualitative analysis of the community and cultural context. *Journal of Pediatric Neurology*, 10(1), 15–22. doi:10.3233/JPN-2012-0527
- Basaffar, A., Niehm, L., & Bosselman, R. (2016). Saudi Arabian women in entrepreneurship: Challenges, opportunities, and potential. In *United States Association for Small Business and Entrepreneurship. Conference Proceedings*. United States Association for Small Business and Entrepreneurship.
- Bellini, S., Peters, J. K., Benner, L., & Hopf, A. (2007). A meta-analysis of school-based social skills interventions for children with autism spectrum disorders. *Remedial and Special Education*, 28(3), 153–162.
- Bos, A., Pryor, J., Reeder, G., & Stutterheim, S. (2013). Stigma: Advances in theory and research. *Basic and Applied Social Psychology*, 35(1), 1–9.
- Busick, M., & Neitzel, J. (2009). *Self-management: Steps for implementation*. Chapel Hill, NC: National Professional Development Center on Autism Spectrum Disorders, Frank Porter Graham Child Development Institute, The University of North Carolina.
- Cappe, E., Wolff, M., Bobet, R., & Adrien, J. (2011). Quality of life: A key variable to consider in the evaluation of adjustment in parents of children with autism spectrum disorders and in the development of relevant support and assistance programmes. *Quality of Life Research: An International Journal of Quality of Life Aspects of Treatment, Care & Rehabilitation*, 20(8), 1279–1294.
- Creswell, J. W. (2007). *Qualitative inquiry and research design: Choosing among five approaches*. Thousand Oaks, CA: Sage.

- Daley, T. (2002). The need for cross-cultural research on pervasive developmental disorders. *Transcultural Psychiatry*, 39, 531–550.
- Davis, N. O., & Carter, A. S. (2008). Parenting stress in mothers and fathers of toddlers with autism spectrum disorders: Associations with child characteristics. *Journal of Autism and Developmental Disorders*, 38, 1278–1291.
- Duarte, C. S., Bordin, I. A., Yazigi, L., & Mooney, J. (2005). Factors associated with stress in mothers of children with autism. *Autism*, 9, 416–427.
- Fernandez, M. C., & Arcia, E. (2004). Disruptive behaviors and maternal responsibility: A complex portrait of stigma, self-blame, and other reactions. *Hispanic Journal of Behavioral Sciences*, 26, 356–372.
- Goffman, E. (1963). *Stigma: Notes on the management of spoiled identity*. Englewood Cliffs, NJ: Prentice Hall.
- Gray, D. (1993). Perceptions of stigma: The parents of autistic children. *Sociology of Health & Illness*, 15(1), 102–120.
- Gurung, R. (2006). *Health psychology: A cultural approach*. Belmont, CA: Wadsworth Publishing.
- Harper, C. B., Symon, J. B., & Frea, W. D. (2008). Recess is time in: Using peers to improve social skills of children with autism. *Journal of Autism and Developmental Disorders*, 38(5), 815–826.
- Hayes, S. A., & Watson, S. L. (2013). The impact of parenting stress: A meta-analysis of studies comparing the experience of parenting stress in parents of children with and without autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 43(3), 629–642. doi:10.1007/s10803-012-1604-y
- Heward, M., Palfreman-Kay, J., & Innes, A. (2015). In their words: How television and visual media can raise awareness of dementia and other health conditions that carry stigma, including disabilities. *The Journal of Popular Television*, 3(2), 229–242.
- Ingersoll, B., & Hambrick, D. Z. (2011). The relationship between the broader autism phenotype, child severity, and stress and depression in

parents of children with autism spectrum disorders. *Research in Autism Spectrum Disorders*, 5(1), 337–344.

Johnson, N., Frenn, M., Feetham, S., & Simpson, P. (2011). Autism spectrum disorder: Parenting stress, family functioning and health-related quality of life. *Families, Systems and Health*, 29(3), 232–252.

Kamps, D. M., Leonard, B. R., Vernon, S., Dugan, E. P., Delquadri, J. C., Gershon, B., & Folk, L. (1992). Teaching social skills to students with autism to increase peer interactions in an integrated first-grade classroom. *Journal of Applied Behavior Analysis*, 25(2), 281–288.

Kohler, F. W., Greteman, C., Raschke, D., & Highnam, C. (2007). Using a buddy skill package to increase the social interactions between a preschooler with autism and her peers. *Topics in Early Childhood Special Education*, 27(3), 155–163.

Kuhn, J. C., & Carter, A. S. (2006). Maternal self-efficacy and associated parenting cognitions among mothers of children with autism. *American Journal of Orthopsychiatry*, 76(4), 564–575.

Lyons, A. M., Leon, S. C., Phelps, C. E. R., & Dunleavy, A. M. (2010). The impact of child symptom severity on stress among parents of children with ASD: The moderating role of coping styles. *Journal of Child and Family Studies*, 19, 516–524.

Mak, W., & Kwok, Y. (2010). Internalization of stigma for parents of children with autism spectrum disorder in Hong Kong. *Social Science & Medicine*, 70, 2045–2051.

Marshall, C., & Rossman, G. B. (1999). *Designing qualitative research*. Thousand Oaks: SAGE Publications.

Martin, D. M. (2012). The ever-changing social perception of autism spectrum disorders in the United States. *Explorations: Social Sciences*, 160–170. Manuscript submitted for publication.

Mash, E., & Wolfe, D. (2010). *Abnormal child psychology* (4th ed.). Belmont, CA: Wadsworth.

McCaslin, M. L., & Scott, L. W. (2003). The five-question method for framing a qualitative research study. *The Qualitative Report*, 8(3), 447–

461.

Miller, V. A., Schreck, K. A., Mulick, J. A., & Butter, E. (2012). Factors related to parents' choices of treatments for their children with autism spectrum disorders. *Research in Autism Spectrum Disorders*, 6(1), 87–95.

Olkin, R. (2002). Could you hold the door for me? Including disability in diversity. *Cultural Diversity and Ethnic Minority Psychology*, 8(2), 130–137.

Olkin, R. (2007). Disability-affirmative therapy and case formulation: A template for understanding disability in a clinical context. *Counseling and Human Development*, 39(8), 1–20.

Patton, M. Q. (2015). *Qualitative research and evaluation methods* (4th ed.). Los Angeles, CA: Sage.

Ravindran, N., & Myers, B. J. (2012). Cultural influences on perceptions of health, illness, and disability: A review and focus on autism. *Journal of Child and Family Studies*, 21, 311–319. doi:10.1007/s10826-011-9477-9

Schulze, M. A. (2016). Self-management strategies to support students with ASD. *Teaching Exceptional Children*, 48(5), 225–231. doi:10.1177/0040059916640759

Solomon, A. H., & Chung, B. (2012). Understanding autism: How family therapists can support parents of children with autism spectrum disorders. *Family Process*, 51(2), 250–264.

van Manen, M. (1990). *Researching lived experiences*. Albany, NY: State University of New York Press.

Wertz, F. J., Charmaz, L., McMullen, L. M., Josselson, R., Anderson, R., & McSpadden, E. (2011). Five ways of doing qualitative analysis: Phenomenological psychology, grounded theory, discourse analysis, narrative research, and intuitive inquiry. New York: Guilford Press.

Zeina, R. M., Al-Ayadhi, L., & Bashir, S. (2014). Autism spectrum disorder: Main problem waiting for solution in Kingdom of Saudi Arabia.

Autism, 8(8), 487–490.



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